

CIE IGCSE BIOLOGY (0610)

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NOTES

CHARACTERISTICS OF LIVING ORGANISMS

MRS GREN

- Movement: action by an organism or part of an organism causing a change of position or place
- Respiration: the chemical reactions that break down nutrient molecules in living cells to release energy
- Sensitivity: ability to detect or sense changes in the environment (stimuli) and to make responses
- Growth: permanent increase in size and dry mass by an increase in cell number or cell size or both
- Reproduction: processes that make more of the same kind of organism
- Excretion: removal from organisms of toxic materials, the waste products of metabolism (chemical reactions in cells including respiration) and substances in excess of requirements
- Nutrition: taking in of nutrients which are organic substances and mineral ions, containing raw materials or energy for growth and tissue repair, absorbing and assimilating them

CLASSIFICATION AND DIVERSITY OF LIVING ORGANISMS

BINOMIAL SYSTEM

KING PHILIP CAME OVER FOR GOOD SPAGHETTI

- Kingdom, Phylum, Class, Order, Family, Genus, Species
- Kingdom → Species = Similarity increases
- Species: organisms which can reproduce successfully
- Binomial system: a system of naming species in which the scientific name of an organism is made up of two parts showing the genus (starting with a capital letter) and species (starting with a lower case letter), written in italics when printed (therefore underlined when handwritten) e.g. *Homo sapiens*

KINGDOMS

- Animal: Multi-cellular ingestive heterotrophs (eat living organisms)
- Plant: Multi-cellular photosynthetic autotrophic (make their own food) organism with a cellulose cell wall.
- Fungi: Single celled or multi cellular heterotrophic organism with a cell wall not made of cellulose, saprotrophs (feed off dead organisms) or parasites
- Monera: Single celled organism with no true nucleus
- Protista: Single celled organism with a nucleus

VERTEBRATES

MR FAB

- Bony fish:
 - Wet scales
 - External fertilization and soft eggs
 - Gills to breathe
- Amphibians:
 - Smooth, moist skin
 - External fertilization and soft eggs
 - Gills/lungs to breathe
 - Can live on land and in water
 - 4 legs
- Birds
 - Feathers on body and scales on legs
 - Have 2 legs and 2 wings
 - Lungs to breathe
 - Hard eggs

- Reptiles:
 - Scales on skin
 - Usually 4 legs
 - Lungs to breathe
 - Hard eggs
- Mammals
 - Fur/hair on skin
 - Can live on land and in water
 - 4 legs
 - Lungs to breathe
 - Give birth to live young

INVERTEBRATES: NO LEGS

MAN

- Molluscs: (e.g. snails)
 - Soft
 - Not segmented - body in three continuous parts
 - Have 1 or 2 shells
 - Moves on muscular foot
 - Mainly herbivore, some carnivores
- Annelids: (e.g. earthworms)
 - Hard, slightly waterproof
 - Rounded bodies
 - Bodies made of segment
 - Uses chaetae (bristles) to move from place to place
 - Mainly herbivores
- Nematode: (e.g. roundworm)
 - Soft, not waterproof
 - Not segmented
 - Wriggles but lives in one place
 - Mainly parasites
 - Circular cross section and tapered ends
 - Invades digestive tracts

ARTHROPODS (INVERTEBRATES WITH LEGS)

CAMI

- Crustaceans: (e.g. crabs)
 - Have an exoskeleton
 - 1 pair of compound eyes
 - 2 body segment – cephalothorax and abdomen
 - More than four pairs of legs
 - 2 pairs of antennae sensitive to touch and chemicals
- Arachnids: (e.g. spiders)
 - 2 body segment – cephalothorax and abdomen
 - Four pairs of legs
 - Pair of chelicerae to hold prey
 - Two pedipalps for reproduction
 - Simple eyes
- Myriapods: (e.g. centipede)
 - Segmented body
 - Additional segments formed
 - One pair of antennae
 - 70+ pairs of legs – 1 or 2 pairs on each segment
 - Fused head and thorax and segmented abdomen
 - Simple eyes
- Insects: (e.g. bees)
 - 3 body segments – head, thorax and abdomen
 - 3 pairs of legs
 - 1 pair of antennae
 - 1 or 2 pairs of wings
 - Compound and simple eyes

VIRUSES AND BACTERIA

	Virus	Bacteria
Covered by	Protein coat	Cell wall
Cell membrane	No	Yes
Cytoplasm	No	Yes
Genetic material	DNA or RNA – only a few genes	DNA – enough for several 100 genes
Living or not?	Non-living unless in host	Living

FUNGI

- Multicellular except for yeast
- Main body is mycelium which is made of hyphae
- Hyphae
 - One cell thick
 - Multiple nuclei
 - Cross for rigidity or no cross walls
- If mycelium beneath soil, produces fruiting body e.g. mushrooms
- Fungi spread by the spreading of spores. Via:
 - Puff balls, pressure trigger
 - Animal carriers
 - Abiotic carriers (wind, water, air)
- The environment needs to be moist, warm, have a nutrient source but light is not necessary, darker environments have less evaporation (so more moist)

MONOCOTYLEDONS

- One cotyledon
- Parallel veins
- Fibrous root
- Floral parts in 3s

DICOTYLEDONS

- Two cotyledons
- Veins netlike
- Taproot present
- Floral parts in 4s or 5s

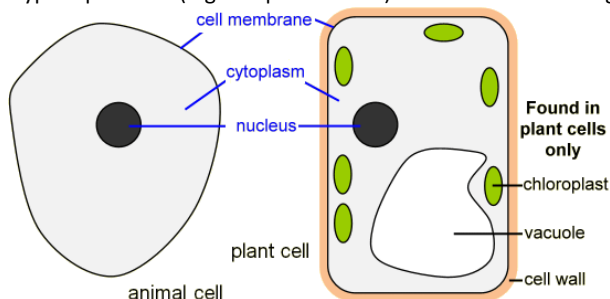
There are other classification systems e.g. cladistics (based on RNA/DNA sequencing data)

SIMPLE KEYS

- **Dichotomous key:** uses visible features to classify organisms. It is which gives you a choice of two features and you follow the one that applies: each choice leads to another choice until the organism is narrowed down to its genus and finally species.

CELL STRUCTURE AND ORGANIZATION

- All living things are made of cells.
- **All typical cells have:**
 - Cell membrane: differentially or partially permeable to allow certain substances to enter and leave the cell.
 - Cytoplasm: where chemical reactions take place
 - Nucleus: contains DNA and controls the cell
 - Mitochondria: organelle where aerobic respiration happens.
- A typical animal cell (e.g. the liver cell) has all the above things.
- **Only plant cells have:**
 - Vacuole: stores food & water & helps to maintain shape of cell
 - Cell wall: rigid to keep shape of cell
 - Chloroplasts: contain chlorophyll, which absorbs light energy for photosynthesis
- A typical plant cell (e.g. the palisade cell) has all the above things.

LEVELS OF ORGANIZATIONSPECIALIZED CELLS

Cell	Function	Adaptation(s)	Diagram
Red blood cell	Transport of oxygen	• Biconcave shape • No nucleus • Flexible • Has hemoglobin	
Muscle cell	Contracts to get structures closer together	• Long • Many protein fibers in cytoplasm to shorten cell when energy available	
Ciliated cell	Move and push mucus	• Tiny hairs called cilia	
Root hair cell	Absorb minerals and water	• Elongated shape for more surface area	
Xylem vessel	Transport water and support plant	• No cytoplasm so water passes freely • No cross walls so cells connect to form tube • Lignin makes it strong and waterproof	
Palisade cell	Photosynthesizes	• Regular shape so many can fit in a small space • Many chloroplasts	

LEVELS OF ORGANIZATION

- **Organelle:** a specialized part of a cell that has its own function, e.g. the nucleus
- **Cell:** the smallest part of a living structure that can operate as an independent unit e.g. red blood cell
- **Tissue:** a group of cells with similar structures, working together to perform a shared function e.g. muscle tissue
- **Organ:** a structure made up of a group of tissues, working together to perform specific functions e.g. the heart
- **Organ system:** group of organs with related functions, working together to perform body functions e.g. respiratory system
- **Organism:** an individual made of organ systems which work to keep that organism alive e.g. a human

SIZE OF SPECIMENS

$$\text{Magnification} = \frac{\text{size of drawing}}{\text{size of specimen}} = \frac{\text{image}}{\text{actual}} = \frac{1}{A}$$

MOVEMENT IN AND OUT OF CELLSDIFFUSION

- This is the movement of molecules from a region of high concentration to a region of low concentration down the concentration gradient.
- This results in random movement of molecules until equilibrium is reached
- The diffusion of gases and solutes is important as without it, molecules which are needed for life, for example glucose and oxygen for respiration, would not be able to get to the places they are needed. Water is needed as a solvent

ACTIVE TRANSPORT

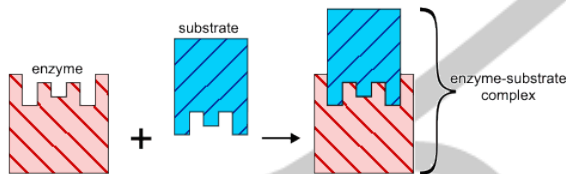
- Movement of ions in or out of a cell through the cell membrane, from a region of lower concentration to a region of higher concentration against a concentration gradient, using energy released during respiration
- Active transport is needed when an organism wants to optimize the amount of nutrients it can take up - ion uptake by root hairs and uptake of glucose by epithelial cells of villi.

OSMOSIS

- Diffusion of water molecules from a region of their higher concentration to a region of their lower concentration, through a partially permeable membrane
- Conc. of solute outside cell = conc. inside cell → no change in size
- Conc. of solute outside cell > conc. inside cell → cell shrinks
- Conc. of solute outside cell < conc. inside cell → cell swells
- In animals:
 - Increasing solute concentration inside of cell can cause cell to explode as a result of it having too much water, crenation.
- In plants:
 - Increasing solute concentration inside of cell causes cell to become turgid, vacuole fills up.
 - Decreasing solute concentration inside of cell causes cell to become flaccid, losing water and vacuole gets smaller. Cell body shrinks, pulling away from the cell wall

ENZYMES

- Catalyst: a substance that speeds up a chemical reaction and is not changed by the reaction
- Enzymes: proteins that function as biological catalysts
- Enzymes lowers amount of energy needed for reaction to take place
- Lock and key theory:



- Substrate: the molecule(s) before they are made to react
- Product: the molecule(s) that are made in a reaction
- Catabolic reaction: molecules are broken down
- Anabolic reaction: molecules are combined

EFFECT OF TEMPERATURE ON ENZYMES

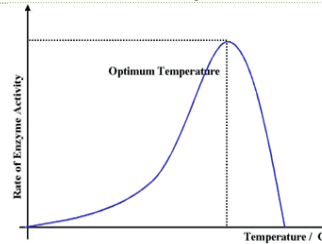
- Enzymes have an optimum temperature: the temperature at which they work best giving the fastest reaction $\approx 37^{\circ}\text{C}$ in animals
- When temperature increases, molecules move faster so collide with an enzyme in less time
- Having more energy makes them more likely to bind to active site.
- If temperature is too high, enzyme molecules vibrate too vigorously and enzyme is denatured; it loses its shape and will no longer bind with a substrate.
- When the temperature is too low there is not enough kinetic energy for the reaction so it reacts too slowly.

EFFECT OF pH ON ENZYMES

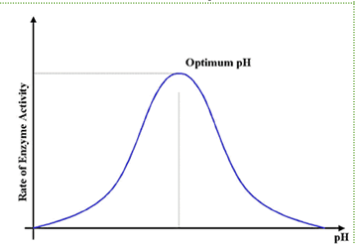
- Enzymes are sensitive to pH
- Some enzymes work best in an acid and others in an alkaline
- Enzymes work best at their optimum pH
- If the pH is changed then the enzyme will denature and will no longer fit with the substrate- no reaction will take place

GRAPHS FOR CHANGES IN ENZYME ACTIVITY

Effect of Temperature



Effect of pH



ENZYMES AND THEIR USES

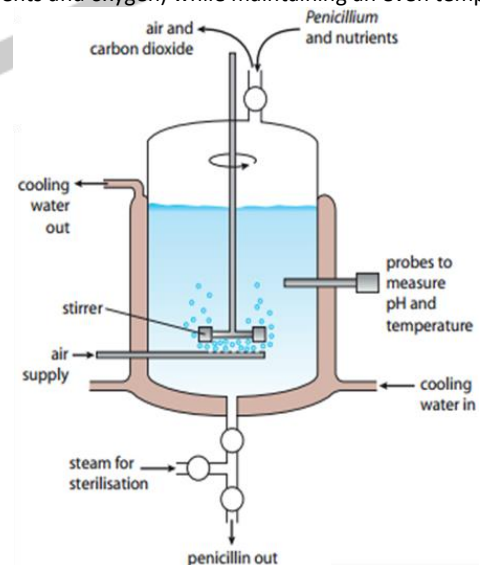
- **Seeds to germinate:** the enzymes turn insoluble food stores to soluble.
- **Biological washing powders:** enzymes are added to washing powders to help remove stains for example:
 - Lipase for lipids from fatty foods and greasy fingerprints
 - Protease for proteins from blood stains
- **Food industry:**
 - Isomerase converts glucose to fructose which is sweeter, so less is needed to give a sweet taste
 - Pectinase helps break down cell walls in fruit juice production so it increases yield, lowers viscosity and reduces cloudiness

MAKING PENICILLIN

- Penicillin: an antibiotic produced by a fungus called penicillium.
- Stainless steel fermentation vessel is filled with medium containing sugars and ammonium salts.
- Penicillium is added to produce penicillin. They use sugar for respiration and ammonium salts to make protein and nucleic acids
- The fermentation vessel consists of

PAWS

- **P**robes monitor temperature and pH
- **A**ir provides oxygen for aerobic respiration in fungus
- **W**ater-cooled jacket removes heat to maintain temperature of 24°C .
- **S**tirrer keeps the microorganism suspended (allowing access to nutrients and oxygen) while maintaining an even temperature.



- Filtered to remove fungus and then can be crystallized to make capsules.

NUTRITION

- **Nutrition:** taking in of nutrients which are organic substances and mineral ions, containing raw materials or energy for growth and tissue repair, absorbing and assimilating them
- **Carbohydrates:** made from Carbon, Hydrogen and Oxygen (CHO)
- **Fats and oils:** made from Carbon, Hydrogen and Oxygen (CHO)
- **Proteins:** made from Carbon, Hydrogen, Oxygen, Nitrogen and sometimes Sulfur (CHON[S])

Basic units (monomers)	Larger molecules (macromolecules)
Simple sugars	Starch and glycogen
Fatty acids and glycerol	Fats and oils
Amino acids	Proteins

CHEMICAL TESTS

- **Starch:** Add few drops of iodine, +ve result = blue-black color
- **Reducing sugars:** Add Benedict's reagent, then mixture is heated in water bath for 2 to 3 minutes. +ve result = brick-red precipitate, -ve result = remains blue
- **Proteins:** Add few drops of Biuret reagent, +ve result = mauve color
- **Fats:** Emulsion test; ethanol is added to mixture, and this is poured into a test tube with an equal amount of distilled water, +ve result = milky-white emulsion

USES

Nutrient	Uses
Carbohydrates	Energy
Fats	Source of energy, building materials, energy store, insulation, buoyancy, making hormones
Proteins	Energy, building materials, enzymes, haemoglobin, structural material (muscle), hormones, antibodies
Vitamin C	Protect cells from ageing, production of fibers
Vitamin D	Absorption of calcium
Calcium	Development and maintenance of strong bones and teeth
Iron	Making haemoglobin
Fiber	Provides bulk for faeces, helps peristalsis
Water	Chemical reactions, solvent for transport

DEFICIENCIES

- **Vitamin C:** *Scurvy*; loss of teeth, pale skin and sunken eyes
- **Vitamin D:** *Rickets*; weak bones and teeth
- **Calcium:** *Rickets*; weak bones and teeth, also poor clotting of blood, spasms
- **Iron:** *Anaemia*; Fatigue (less iron → less haemoglobin → less oxygen transported → less respiration → less energy)

FOOD ADDITIVES

- Substances with no nutrient value which are added to improve appearance, flavor, texture and/or storage properties of food
- Preservatives inhibit growth of fungi/bacteria e.g. SO₂ to control browning of potatoes and anti-oxidants prevent deterioration.
- Food additives can have health hazards for example sulfur dioxide causes sensitivity in asthma sufferers.

SINGLE-CELL PROTEIN

- Sources of mixed protein extracted from pure or mixed cultures of algae, yeasts, fungi or bacteria (grown on agricultural wastes)
- Used as a substitute for protein-rich foods, for human & animal
- Excess yeast from alcoholic fermentation is sometimes used as cattle feed.

YOGHURT

- Soured milk, partially clotted, with a mildly acidic taste.
- In incubation, culture of bacteria are kept at 45°C and turn lactose into lactic acid during respiration, then cooling at 4°C stops reaction

PLANT NUTRITION

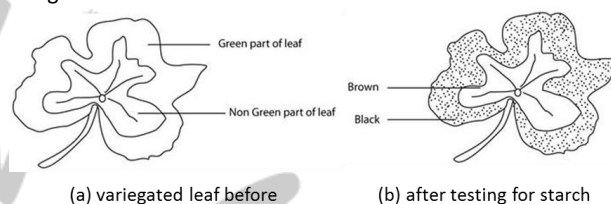
- **Photosynthesis:** fundamental process by which plants manufacture carbohydrates from raw materials using energy from light.

$$\text{Carbon Dioxide} + \text{Water} \xrightarrow{\text{light + chlorophyll}} \text{Glucose} + \text{Oxygen}$$

$$6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{light + chlorophyll}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$$
- The carbon dioxide diffuses through the open stomata of the leaf of a plant and water is taken up through the roots.
- Chlorophyll is a dye, which traps light energy and converts it into chemical energy for the formation of carbohydrates and their subsequent storage

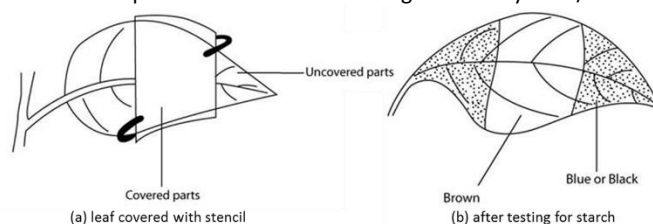
CHLOROPHYLL IS NECESSARY FOR PHOTOSYNTHESIS

- Take a potted plant with variegated (green and white) leaves.
- Destarch the plant by keeping it in complete darkness for about 48 hours.
- Expose the plant to the sunlight for a few days.
- Leaf boiled in water for 2 minutes to break down cell walls, denature enzymes and allow for easier penetration by ethanol.
- Warmed in ethanol until leaf is colorless to extract chlorophyll, which would mask observation
- Dipped in water briefly: to soften leaf
- Leaf is placed on a white tile and iodine is added. If starch is present, color will be blue-black and if absent, it will remain orange



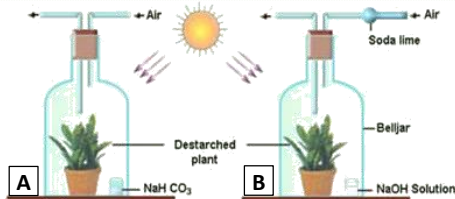
LIGHT IS NECESSARY FOR PHOTOSYNTHESIS

- Destarch the plant by keeping it in darkness for 48hrs
- Place a stencil over part of a leaf
- Place the leaf in sunlight for 4-6 hours
- Remove the stencil and test for starch
- +ve result = parts which received light turn black
- -ve result = parts which didn't receive light remain yellow/brown



CARBON DIOXIDE IS NECESSARY FOR PHOTOSYNTHESIS

- Take two destarched potted plants.
- Cover both the plants with bell jars and label them as A and B.
- Inside A, keep NaHCO₃ (sodium bicarbonate). It produces CO₂.
- Inside B, keep NaOH (Sodium hydroxide). It absorbs CO₂.
- Keep both the set-ups in the sunlight for at least 6 hours.
- Perform the starch test on both of the plants.



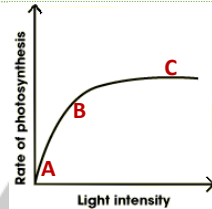
- The leaves of Plant A will turn black after the starch test
- The leaves of Plant B will remain orange/brown after starch test

LIMITING FACTORS

- Limiting factor: something present in the environment in such short supply that it restricts life processes.

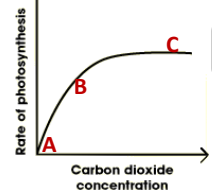
Light Intensity

- As the amount of light increases, the rate of photosynthesis increases (A-B)
- The limiting factor is light
- Increasing the amount of light after a certain point has no effect on the rate (C)
- The limiting factor is now carbon dioxide or temperature



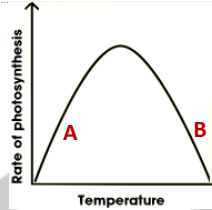
Carbon Dioxide Concentration

- As the amount of carbon dioxide increases, the rate of photosynthesis increases (A-B)
- The limiting factor is Carbon dioxide
- Increasing amount of Carbon dioxide after a certain point has not effect on rate (C)
- The limiting factor is now light or temperature (warmth)



Temperature

- As temperature increases, the rate of photosynthesis increases until it reaches optimum temperature 40°C (A)
- The limiting factor is the temperature
- Increasing the temperature above 40°C will cause the enzymes to denature (B)
- This will decrease rate of photosynthesis

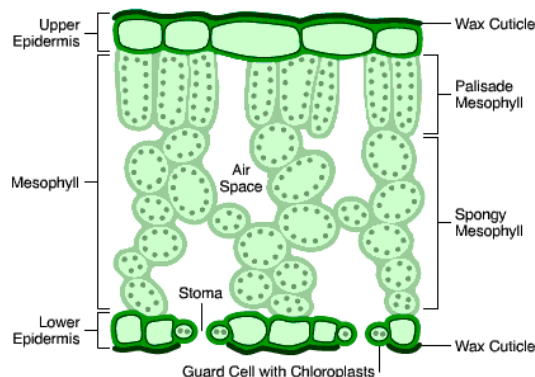


GLASSHOUSE SYSTEMS

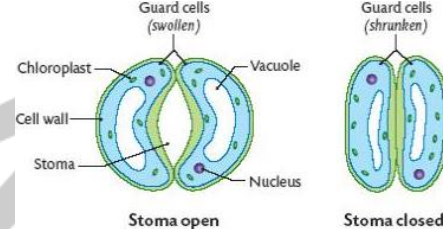
To increase the crop yield, farmers control the limiting factors:

- CO₂ enrichment: paraffin is burnt to increase CO₂ concentration by three times the original amount and doubling the yield
- Optimum temperature: thermostatically controlled heaters make the temperature right for the enzymes to work
- Optimum light: light has a high intensity for more photosynthesis, the correct wavelengths (red and blue not green) and duration controls production of fruit

LEAF STRUCTURE

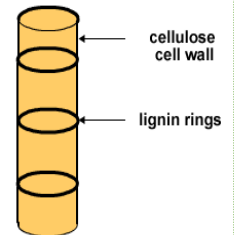


- **Cuticle:** waxy layer that prevents water loss from top of the leaf
- **Epidermis:** transparent cell that allows sunlight to pass through to the palisade cell
- **Palisade:** found at the top of the cell and contains many chloroplasts which absorb sunlight.
- **Spongy mesophyll layer:** irregularly shaped cells which create air spaces to allow gaseous exchange to take place; do not contain many chloroplasts
- **Vascular Bundle:** made up of xylem and phloem
- **Xylem:** vessel which transports water and dissolved minerals and has lignified walls made of cellulose
- **Phloem:** vessel which transports nutrients
- **Stomata:** little holes that open and close to allow gaseous exchange to take place. The stomata close to prevent water loss and open to let gases come in and out. When guard cells lose water, the stomata close (at night), while the stomata open when guard cells gain water & swell (during the day).



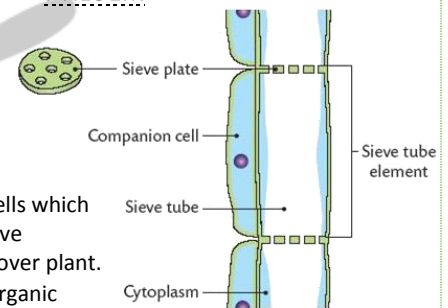
XYLEM

- Unidirectional vessel which transports water and dissolved minerals.
- Walls are made out of waterproof lignin.
- Water moves up due to transpiration and osmosis



PHLOEM

- Bidirectional vessel
- Contains sieve elements which allow sugars to pass from one cell to next downwards
- Contains companion cells which provide energy for active transport of sugars all over plant.
- Translocation moves organic molecules (sugars, amino acids) from source to sink.
- Phloem vessels still have cross walls called sieve plates that contain pores.
- Companion cells actively load sucrose into the phloem.
- Water follows high solute in phloem by osmosis. A positive pressure potential develops moving mass of phloem sap forward.
- Phloem still contains small amount of cytoplasm along the walls but the organelle content is greatly reduced.
- Companion cells actively unload (ATP used) the organic molecules



MINERAL REQUIREMENTS

Nitrogen

- Needed for protein synthesis
- Deficiency: small plant, slow growth, top leaves pale, bottom leaves dead and roots slightly affected

Magnesium

- Needed for chlorophyll synthesis
- Deficiency: plant lack chlorophyll, leaves yellow but normal roots

- **Nitrogen fertilizers:** provide nitrogen in the form of nitrate ions, nitrite ions or ammonium ions. But using fertilisers can lead to eutrophication, which is when the fertiliser is transported by rain and leaches into stagnant water e.g. pond or river

DIET

- **Balanced Diet:** getting all the right nutrients in correct proportions
- Diet related to age/sex/activity:
 - **Children Below 12:** Require more calcium
 - **Teenagers:** Highest calorie Intake
 - **Adults:** Balanced meal with less calories
 - **Pregnant Women:** more iron, calcium and folic acid
 - **Males:** Generally require more energy

MALNUTRITION

- A condition caused by eating an unbalanced diet. Several forms:
 - **Overnutrition:** balanced diet but eating too much of everything
 - **Undernutrition:** having too little food
 - **Eating foods in incorrect proportions**

EFFECTS OF MALNUTRITION

- **Starvation:** losing strength & finally dying because of lack of food
- **Coronary heart disease:** eating too much fats which are rich in saturated fatty acids and cholesterol, may lead to heart attack
- **Constipation:** lack of roughages in food causes constipation because roughages are indigestible and form bulks. Friction between bulks and walls of intestine stimulate the peristalsis
- **Obesity:** Eating too much fats and carbohydrates leads to their storage in the body mainly in the forms of fats and causing an increase in body weight. This can cause; heart attack, stroke, joint pain, mobility impairment, high blood pressure

FOOD SUPPLY

Food production has increased because:

- **Improved machinery** means less labor is needed
- **Fertilizers** help crops to grow better
- **Insecticides:** a type of pesticide that kills insects
- **Herbicides:** a type of pesticide that kills weeds
- **Artificial selection and genetic modification** means that yields are improved: cows produce more milk, cows are more muscular giving more meat, plant crops can resist insects and cold weather

WORLD FOOD SUPPLIES

- Not enough food available in a country to feed its people because:
 - Fast increase in population
 - Increasing use of crops for fuel
 - Decrease of farming = Climate change/Urbanization
- **Famine:** Wide spread scarcity of food
- The main causes of famine:
 - The rapid rate of population increase
 - Long term climatic change
 - Soil erosion and desertification
 - Economic pressure
 - Unequal distribution of food
 - Drought
 - Flood

HUMAN ALIMENTARY CANAL

- **Ingestion:** taking substances (e.g. food, drink) into the body through the mouth.
- **Egestion:** passing out of food that has not been digested, as faeces, through the anus.
- **Digestion:** the break-down of large, insoluble food molecules into small, water soluble molecules using mechanical and chemical processes

- **Mouth:** contains teeth used for mechanical digestion, area where food is mixed with salivary amylase & where ingestion takes place
- **Salivary glands:** produce saliva which contains amylase and helps food slide down oesophagus

- **Oesophagus:** tube-shaped organ which uses peristalsis to transport food from mouth to stomach

- **Stomach:** has sphincters to control movement into and also has pepsin (a protease) to break down proteins into peptides, it also kills bacteria with hydrochloric acid. They also have elastic walls.

- **Small intestine:** tube shaped organ composed of two parts the:

- **Duodenum:** fats are emulsified by bile, and digested by pancreatic lipase to form fatty acids and glycerol. Pancreatic amylase and trypsin (a protease) break down starch and peptides into maltose and amino acids
- **Ileum:** Maltase breaks down maltose to glucose. This is where absorption takes place; adapted by having villi and microvilli.

- **Pancreas:** produces pancreatic juice which contains amylase, trypsin and lipase and hydrogencarbonate.
- **Liver:** produces bile, stores glucose as glycogen, interconverting them to keep glucose concentration constant. Also carries out interconversion of amino acids (transamination), deamination and removal of old red blood cells and storage of their iron. Also site of breakdown of alcohol and other toxins.

- **Gall bladder:** stores bile from liver

- **Bile:** produced by liver and stored in gall bladder, its role is to emulsify fats, to increase surface area for the action of enzymes.

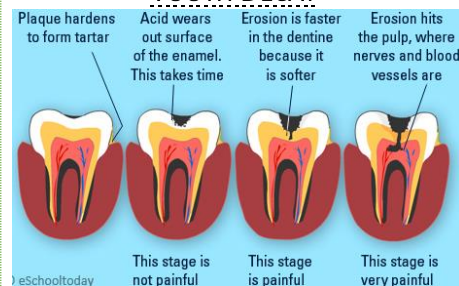
- **Large intestine:** tube shaped organ composed of two parts:
 - **Colon:** organ for absorption of minerals and vitamins, and reabsorbing water from waste to maintain body's water levels
 - **Rectum:** where faeces are temporarily stored

- **Anus:** ring of muscle which controls when faeces is released.

TEETH

Incisor	Canine	Premolar	Molar
			
Rectangular shape, sharp for cutting and biting	Sharp-pointed for holding and cutting	Blunt for chewing and crushing	Blunt chewing and crushing. Two roots

TOOTH DECAY

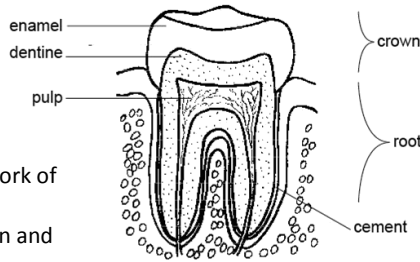


PREVENTION

- Eating food with low sugar content
- Regular and effective teeth brushing to remove plaque
- Finishing a meal with a crisp vegetable and a glass of water

STRUCTURE OF A TOOTH

- **Enamel:** the strongest tissue in the body made from calcium salts
- **Cement:** helps to anchor tooth
- **Pulp cavity:** contains tooth-producing cells, blood vessels, and nerve endings which detect pain.
- **Dentine:** calcium salts deposited on a framework of collagen fibers
- **Neck:** in between crown and root, it is the gums



FLUORIDATION

- Helps teeth by:
 - promoting tooth remineralisation by attracting other minerals like calcium
 - It helps to make the tooth decay-resistant
 - Slows down production of acids by bacteria
- Addition of fluoride to public water supplies:

For

- Helps to strengthen tooth enamel
- Available to all
- Treats whole population
- Free (to people)
- Cheap to supply

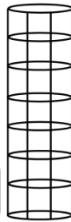
Against

- Allergies such as gastric disturbance, cardiovascular problems, head ache, fits
- Bad taste
- Dosage not controlled for individuals
- No individual choice
- Discoloured teeth - fluorosis

- **Mastication (chewing):** to grind up food or other material with the action of the teeth and jaws

PERISTALSIS

- Waves of involuntary muscle contractions that transport food, waste matter, or other contents through a tube-shaped organ such as the intestine.
- The organ contains circular muscles (rings) and longitudinal muscles (lines).
- Circular muscles contract on either side of the bolus to push it downwards but not letting it fall.
- Longitudinal muscles contract to shorten the organ.



CHEMICAL DIGESTION

- Where enzymes are used to break down large insoluble substances such as proteins into smaller soluble substances like amino acids so that they can be absorbed.
- **Amylase:** breaks down starch into maltose, it is produced in the pancreas (but also in the salivary gland)
- **Protease:** breaks down proteins to peptides (done by pepsin) then into amino acids (done by trypsin). Pepsin comes from the stomach and trypsin comes from the pancreas.
- **Lipase:** breaks down lipids into fatty acids and glycerol, produced by the pancreas.
- Refer to Annex-1 for detailed table

ABSORPTION

- Movement of digested food molecules through wall of the intestine into the blood or lymph.
- The small intestine is the region for absorption of digested food.

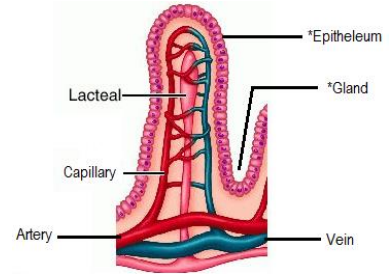
- The small intestine is folded into many villi which increase the surface area for absorption. One villus will have tiny folds on the cells on its outside called microvilli.
- More surface area means more absorption can happen.

- **Capillary:** transports glucose and amino acids

- **Vein:** delivers absorbed products to liver via hepatic portal vein.

- **Gland:** produces enzymes

- **Epithelium:** only one cell thick for faster transport. The cells of the epithelium are folded to form microvilli.



- Small intestine and colon absorb water
 - The small intestine absorbs 5–10 dm³ per day
 - The colon absorbs 0.3–0.5 dm³ per day

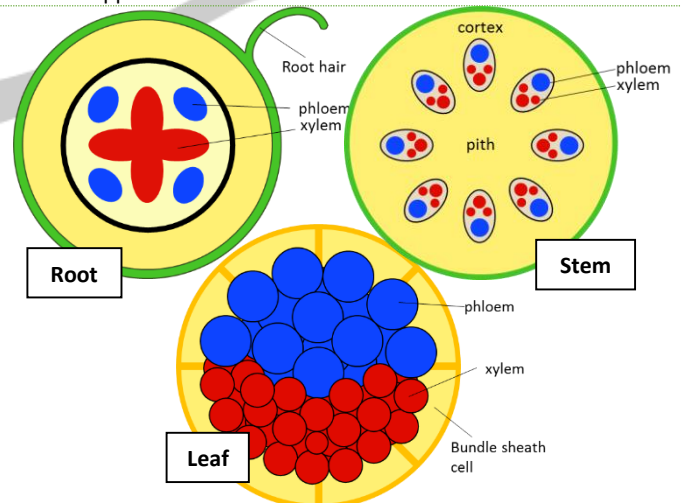
ASSIMILATION

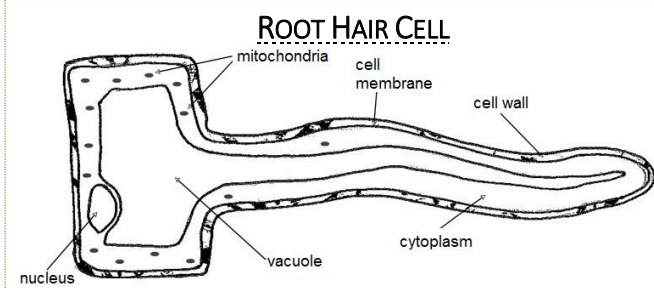
- Movement of digested food molecules into cells of the body where they are used, becoming part of the cells.
- The liver releases nutrients in ideal concentrations through the hepatic vein to tissues around the body.
- **The liver in metabolism:**
 - Converts glucose into glycogen as a means of storage (because glycogen is insoluble)
 - Converts amino acids into proteins, and destroys the excess
- Fat is an energy storage substance.
- **Deamination:** removal of nitrogen from amino acids to form urea, followed by release of energy from remainder of the amino acid.
- Liver is also the site of breakdown of alcohol and other toxins.

TRANSPORTATION

TRANSPORT IN PLANTS

- Functions of xylem and phloem
 - To transport substances from source, where they are taken in or made, to the sinks, where they are used
 - To support the stem

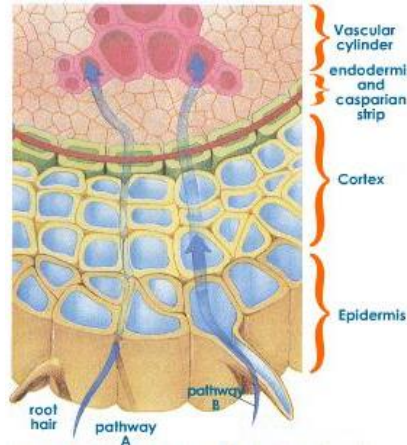




- Function: to absorb water and minerals from the soil.
- They have an elongated shape for more surface area.

PATHWAY TAKEN BY WATER

- Water enters root hair cell from moist soil via osmosis because water potential is higher in soil than in cytoplasm.
- Water passes through the cortex cells by osmosis but mostly by "suction".
- Water and minerals are forced to cross the endodermis.
- Water enters xylem then leaves when it gets to mesophyll cells



TRANSPIRATION

- Evaporation of water at surfaces of the mesophyll cells followed by loss of water vapor from plant leaves, through stomata.
- Water leaves mesophyll cells, into air spaces created by irregular shape of spongy mesophyll cells, then diffuses out of the stomata.
- **Wilting:** occurs if water loss is greater than water uptake – cells become flaccid, tissues become limp and plant no longer supported

UPTAKE OF WATER

- Caused by water loss in leaves which lowers its water potential
- Water moves from xylem to enter leaf tissues via osmosis
- Water moves up the stem in the xylem due to tension (because of cohesion of water molecules to each other) caused by water loss from the leaves
- Ends with the gain of water through roots
- This upward flow of water is called the transpiration stream.

FACTORS AFFECTING RATE OF TRANSPIRATION

- **Temperature:** higher temperatures increase water-holding capacity of air and increases transpiration rate
- **Humidity:** low humidity increases water potential gradient between leaf and atmosphere $\therefore$ increasing transpiration rate
- **Light intensity:** high light intensity causes stomata to open (to allow more photosynthesis) which causes increase in transpiration

PLANT ADAPTATION IN CONTRASTING ENVIRONMENTS

- **Desert:**
 - Cacti has green stem which carries out photosynthesis
 - Small spiky leaves to reduce surface area for water loss
 - Stomata sunk in grooves to avoid drying winds
 - Swollen stem stores water
 - Shallow roots absorb lightest rainfall
 - Deep roots penetrate to very low water table.

• Pond:

- Little lignin in xylem, since leaf is supported by water
- Very thin cuticle since water is plentiful
- Stomata on upper surface, allow CO_2 uptake from atmosphere

• Garden:

- Leaves collapse and stomata close to reduce heat absorption and transpiration of water
- In severe conditions, plants allow leaves to fall off so no water loss can occur. No photosynthesis takes place, but plants can remove chlorophyll from leaves for storage (yellow/red leaves)

TRANSLOCATION

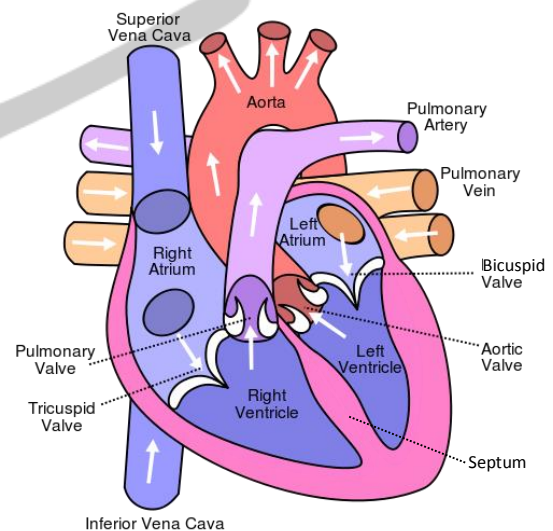
- Movement of sucrose and amino acids in phloem; from regions of production (sources) to regions of storage or to regions of utilization in respiration or growth (sinks).
- Translocation in different seasons:
 - Spring: sucrose transported from stores in roots to leaves
 - Summer & early autumn: sucrose goes from photosynthesizing leaves to root stores,
- Systemic pesticides:
 - Aphids (greenfly) insert mouthpiece into phloem to take nutrients.
 - Systemic insecticides are sprayed onto plants and absorbed into phloem
 - Used to kill only pests (the aphids) instead of killing useful insect species (pollinators)

TRANSPORT IN HUMANS

- **Circulatory system:** system of tubes (veins, capillaries, arteries) with a pump (heart) and valves (in heart and veins) to ensure one-way flow of blood.
- **Double circulation system:**

<u>Pulmonary</u>	<u>Systemic</u>
• Through the lungs	• Through the whole body
• Weaker circulation	• Stronger circulation
• Less pressure	• More pressure

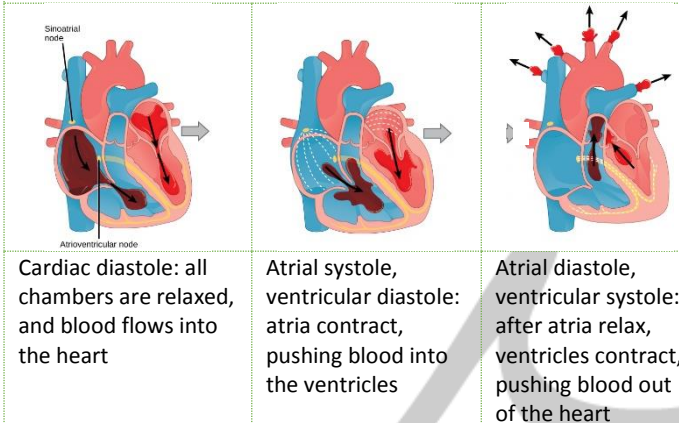
HEART



- **Right atrium:** collect deoxygenated blood & pump it to right ventricle
- **Right ventricle:** pumps deoxygenated blood to lungs
- **Pulmonary artery:** carries deoxygenated blood from right ventricle to lungs
- **Septum:** separates left and right sides of the heart
- **Pulmonary vein:** carry oxygenated blood from lungs to left atrium

- **Left atrium:** collect oxygenated blood and pump it to left ventricle
- **Left ventricle:** pumps oxygenated blood to the body via the aorta
- **Aorta:** carries oxygenated blood from left ventricle to rest of body
- **Tricuspid and bicuspid valves:** prevent backflow of blood into the atria when ventricles contract (atria ventricular valves)
- **Pulmonary and aortic valves:** prevent backflow of blood from the arteries into the ventricles (semi-lunar valves)

CARDIAC CYCLE



- Physical activity makes the heart beat more quickly and more deeply, for an increased circulation of blood so that more oxygen and glucose can get to the muscles.

CORONARY HEART DISEASE

- Coronary artery becomes blocked, interrupting the supply of blood to the heart muscle.
- The heart muscle cells are deprived of oxygen and glucose, and poisonous wastes such as lactic acid build up.
- Part of the heart muscle stops contracting, causing a heart attack
- Caused by stress, smoking, poor diet, poor lifestyle & genetically
- Can be prevented by not smoking, avoiding fatty food and exercising regularly

ARTERIES, VEINS AND CAPILLARIES

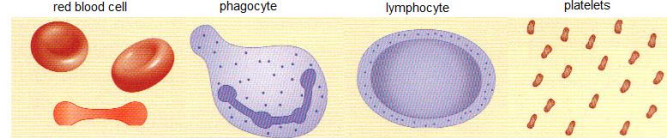
Lungs→heart = Pulmonary vein	Heart→lungs = Pulmonary artery
Liver→heart = Hepatic vein	Heart→liver = Hepatic artery
Kidneys→heart = Renal vein	Heart→kidneys = Renal artery

Vessel	Function	Structure
Artery	• Transport high pressure blood away from heart	• Elastic walls expand and relax as blood is forced out; causes pulse • Thick walls withstand high pressure • Rings of muscle narrow or widen artery to control blood flow.
Vein	• Transport low pressure blood to the heart	• Valves prevent backflow of blood. • Blood is at low pressure, but nearby muscles squeeze veins and help push blood to the heart • Large diameter and thin walls reduce resistance to flow of blood
Capillary	• Allow substances to diffuse into cells	• One cell thick walls for easy diffusion • Highly branched; large surface area • Capillary beds constantly supplied with fresh blood, so diffusion occurs

- Useful substances move out of plasma of capillaries into tissue fluid (fluid in between cells in tissues)
- Cells need oxygen and nutrients, and produce waste products such as CO₂ and useful products such as hormones.
- Capillaries are constantly supplied with new blood, otherwise diffusion could not occur

BLOOD

- Red blood cells: hemoglobin and oxygen transport
- White blood cells: phagocytosis and antibody formation
- Platelets: causing clotting
- Plasma: transport of blood cells, ions, soluble nutrients, hormones, carbon dioxide, urea and plasma proteins



IMMUNE SYSTEM

- Phagocyte has lobed nucleus and vesicles containing digestive enzymes.
- Phagocytosis: engulf pathogen, vesicles fuse with vacuole, enzymes digest bacteria.
- Antigen: protein/carbohydrate on surface of pathogen which provokes immune system.
- Lymphocytes are found in blood and in lymph nodes
- Large nucleus and they produce antibodies,
- Antibodies: Y-shaped protein, bind to label pathogens.
- Then either destroyed by being ingested by phagocytes, or the antibodies may do it.

LYMPHATIC SYSTEM

- Circulation of body fluids, and the production of lymphocytes.
- Lymph node contains many lymphocytes which filter lymph.
- Tissue fluid: made when plasma is squeezed out of capillaries.
- Substances diffuse between cells and tissue fluid.
- Lymph vessels collect lymph and return it to the blood.
- Tissue fluid returns to the capillaries by osmosis.

BLOOD CLOTTING

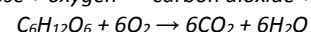
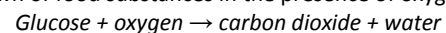
- Reduces blood loss and keeps pathogens out
- Fibrinogen (inactive) turns to fibrin (activated), and forms a mesh to trap red blood cells, which eventually dries to form a scab.

RESPIRATION

- Chemical reactions that break down nutrient molecules in living cells to release energy.
- Uses of energy in the body of humans: muscle contraction, protein synthesis, cell division, active transport, growth, the passage of nerve impulses and the maintenance of a constant body temperature.

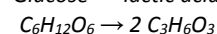
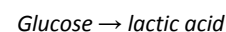
AEROBIC RESPIRATION

- Release of a relatively large amount of energy in cells by the breakdown of food substances in the presence of oxygen.



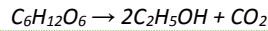
ANAEROBIC RESPIRATION

- Release of a relatively small amount of energy by the breakdown of food substances in the absence of oxygen.
- In muscles:



• **In yeast** (single-cell fungi):

Glucose $\rightarrow$ ethanol + carbon dioxide



USES OF ANAEROBIC RESPIRATION

Brewing

- Grapes (sugar source) are pressed to allow enzymes to begin fermentation
- Yeast converts sugar into alcohol.
- At 8-9% the alcohol (which is toxic) kills the yeast
- Higher concentrations are achieved by distillation

Bread Making

- Flour, sugar, water and salt are mixed with yeast to make the dough.
- The dough is kept in a warm, moist environment (28°C). Yeast ferments sugar making carbon dioxide which creates bubbles, so bread rises
- Cooking (at 180°C) – kills yeast, evaporates alcohol and hardens outer surface.

• **Disadvantages of anaerobic respiration:**

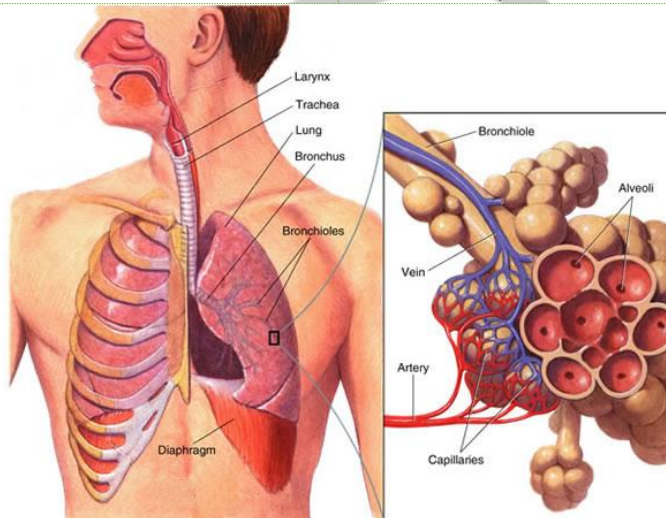
- Only produces 1/20 of the energy per glucose molecule that aerobic respiration would
- Produces poisonous lactic acid

• **Lactic acid:**

- Transported in blood to heart, liver and kidneys, which oxidize it
- The heart, liver and kidneys need extra oxygen to do this which causes you to continue breathing heavily after exercise.
- The extra oxygen is called the oxygen debt.

GAS EXCHANGE

Property of surface	Reason
Thin (ideally one cell thick)	Short distance to diffuse
Large surface area	Many molecules can diffuse at once
Moist	Cells die if not kept moist
	Concentration gradients for oxygen and carbon dioxide are kept up by regular fresh supplies of air
Well ventilated	
	Gases can be carried to/from the cells that need/produce them



Inspired Air

- 21% oxygen
- 0.04% carbon dioxide
- 78% nitrogen
- Water vapour varies to climate

Expired Air

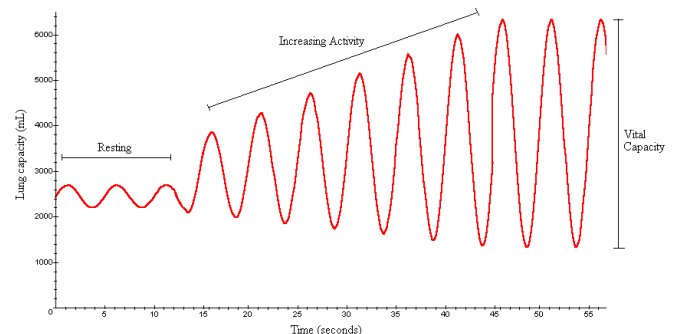
- 18% oxygen
- 3% carbon dioxide
- 78% nitrogen
- Saturated water vapour.

- **Test for CO₂:** Blow CO₂ through limewater. +ve result = turn cloudy

EFFECT OF PHYSICAL ACTIVITY ON BREATHING

- Physical activity increases the breathing rate – more breaths per minute, and the tidal volume – more air per breath
- This is measured with a spirometer to produce a spirogram.
- During exercise, tissues respire at a higher rate, the change in breathing volume and rate helps to keep CO₂ concentration and pH at safe levels.

Spirogram:

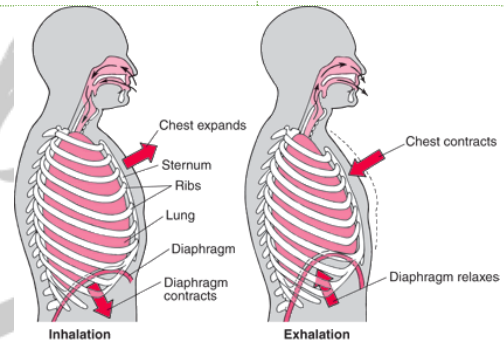


Breathing In

- External intercostal muscles contract – pulls rib cage upwards and outwards
- Diaphragm muscles contract – diaphragm moves upwards
- Lung volume increases – and pressure falls
- Air rushes in to equalise pressure

Breathing Out

- External intercostal muscles relax – rib cage falls downwards and inwards
- Diaphragm muscles relax – returns to dome shape
- Lung volume decreases – and pressure increases
- Air is forced out



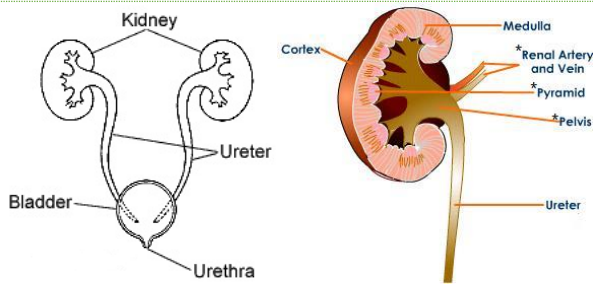
- **Internal intercostal muscles:** are used in coughing and sneezing.
- **Mucus & cilia:** goblet cells produce sticky mucus to trap and eliminate particulate matter and microorganisms.
- **Ciliated cells** have cilia: little hairs which sweep back and forward in a coordinated way to brush mucus up the lungs into the mouth

EXCRETION IN HUMANS

- **Excretion:** the removal from organisms of toxic materials, the waste products of metabolism (chemical reactions in cells including respiration) and substances in excess of requirements.
- Substances should include carbon dioxide, urea and salts.

FUNCTION OF KIDNEY

- Removal of urea and excess water and the re-absorption of glucose and some salts
- Urea is formed in the liver from excess amino acids
- Alcohol, drugs and hormones are broken down in the liver.



DIALYSIS

- When a kidney machine takes a patient's blood and cleans it, then returns the blood to circulation.
- This is how it works:
 - Blood enters machine from patient
 - The pump passes the blood through the dialysis tubing which is semi-permeable therefore acting as a filter
 - The surrounding liquid contains some salts, glucose but no urea so waste materials pass from blood by diffusion;
 - The 'cleaned' blood returns to patient's circulation/body

- **Cortex:** contains Bowman's capsules and coiled tubules
- **Ureter:** carries urine from kidney to bladder
- **Medulla:** contains loops of Henlé and collecting ducts
- **Loop of Henlé:** selectively absorbs water/solutes
- **Collecting ducts:** reabsorbs water into blood and store wastes until they are passed into ureter
- **Urethra:** carries urine from bladder to the outside.
- **Bladder:** stores urine
- **Renal capsule:** filters from blood: water, glucose, urea and salts.
- **Tubule:** (yellow) reabsorbs 100% of glucose, most of the water and some salts back into the blood (red), leading to concentration of urea in the urine as well as loss of excess water and salts into the tubule.
- **Renal artery:** brings wastes and water from blood
- **Renal vein:** reabsorbs water and useful molecules and leaves wastes behind

Dialysis

- More expensive in the long run
- Very disruptive (three 6-8 hrs sessions per week)
- Do not need to find kidney
- Need a machine & must live near one

Transplant

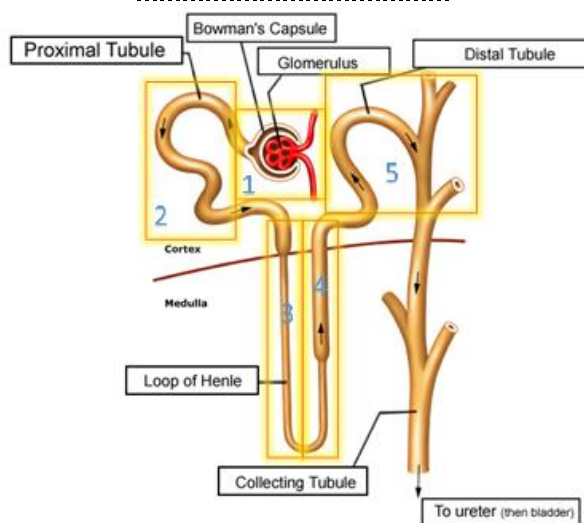
- Less expensive in the long run
- Not very disruptive (only have to take medication)
- Need a kidney
- Can go anywhere, anytime
- Risk of rejection

COORDINATION AND RESPONSE

NERVOUS CONTROL IN HUMANS

- The nervous system consists of two parts:
 - **Central nervous system (CNS)** consisting of the brain and spinal cord, which are the areas of coordination
 - **Peripheral nervous system (PNS)** made up of nerves and neurones, which coordinate and regulate bodily functions.

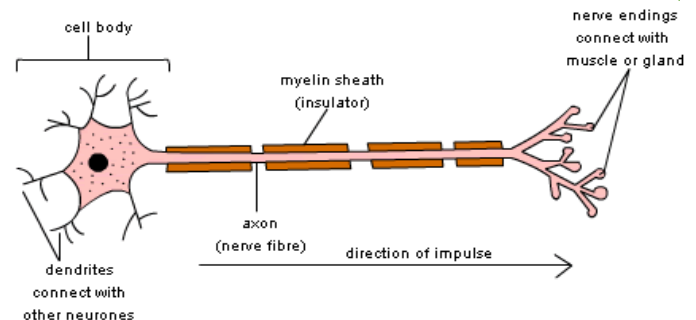
STRUCTURE OF THE KIDNEY



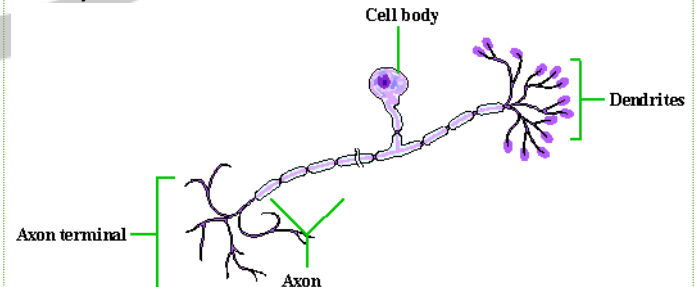
1. **Ultrafiltration:** blood from renal artery enters the glomerulus. Water, urea, salts and glucose are forced into the Bowman's capsule. Blood cells and large proteins cannot pass through.
2. **Selective reabsorption:** in the proximal tubule two thirds of the salt and water and all the glucose moves out of the nephron, by active transport. These substances are reabsorbed back into the blood capillary.
3. **Loop of Henlé:** this part of the loop of Henlé is permeable to water but not salt. Water passively diffuses out of the nephron because of the low water potential of the medulla tissue fluid.
4. **Loop of Henlé:** this part is permeable to salt but not water. The loss of water from the filtrate in the previous stage increases the salt concentration. Some salt passively diffuses out of the loop here.
5. **Collecting duct:** the remaining substances move through the second coiled tubule (distal tubule), into the collecting duct. The permeability of this part of the nephron to water is controlled

TYPES OF NEURONS

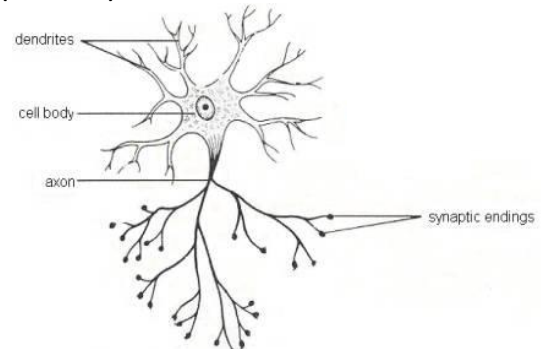
Motor Neurone:



Sensory Neurone:

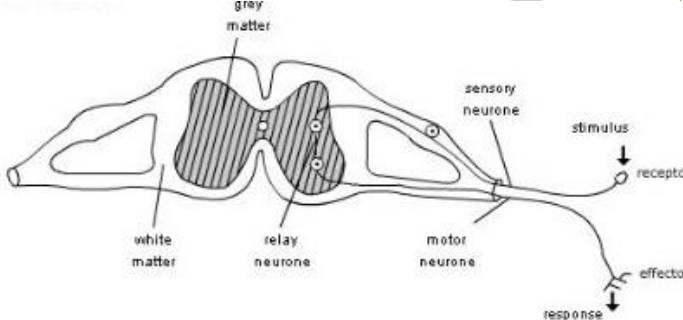


Relay (connector) neurone:



REFLEX ARC

- A reflex action is an involuntary, quick action to respond to a stimulus, in order to protect the body from danger
- E.g. quickly removing your hand from a hot metal surface.
- They involve three neurones: a sensory neurone, relay neurone and motor neurone.
- The gap between neurones is called a synapse.
- **How the reflex arc works:**
 - A stimulus affects a receptor (cell or organ that converts a stimulus into an electrical impulse)
 - A sensory neurone carries impulse from the receptor to the CNS
 - Connector/relay neurone carries impulse slowly (because it has no myelin sheath) across the spinal cord
 - Motor neurone carries impulse from the CNS to the effector
 - Effector (either a muscle or a gland) carries out the response

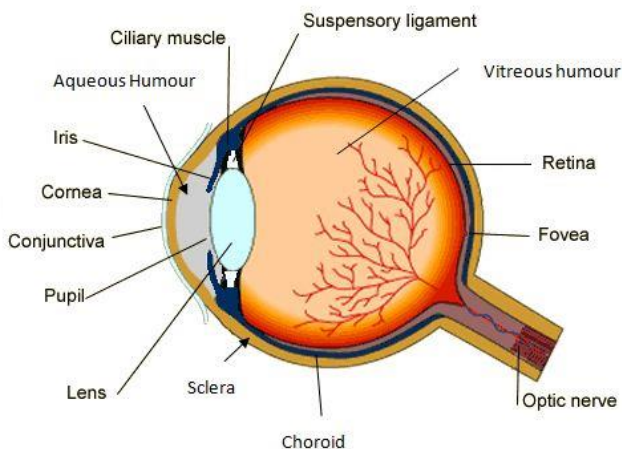


ANTAGONISTIC MUSCLE

- A muscle that opposes the action of another; e.g. biceps and triceps are antagonistic muscles or circular and radial muscles in the eye
- Agonist: a muscle that contracts while another relaxes; e.g. when bending the elbow the biceps are the agonist
- Antagonist: a muscle that relaxes while another contracts; e.g. when bending the elbow the triceps are the antagonist
- **Sense organ:** groups of receptor cells responding to specific stimuli: light, sound, touch, temperature and chemicals.

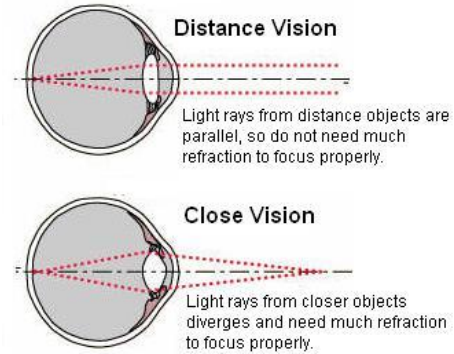
THE EYE

- The sense organ responsible for sight



ACCOMODATION

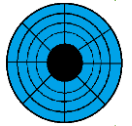
- Adjusting for near and distant objects.



Near Object	Distant Object
• Ciliary muscles contract • Ligaments relax • Lens becomes short and fat	• Ciliary muscles relax • Ligaments are tight • Lens becomes long and thin

PUPIL REFLEX

- Adjusting for high and low light intensity



Low Light Intensity	High Light Intensity
• Radial muscles (straight lines) contract and become shorter to pull the pupil (black dot) making it wider, to let more light enter, to form a clear image on retina	• Circular muscles (circular lines) contract and become shorter to reduce the size of the pupil to protect retina from bleaching.

RODS AND CONES

Rods	Cones
• Provide low detail, black & white images, good for seeing in low intensity light (at night). • Packed most tightly around edge of retina so you can see things most clearly when not looking directly at them.	• Provide detailed, coloured images; they work in high light intensity. • Most tightly packed at centre of retina, so objects are seen most clearly when being directly looked at.

HORMONES

- A chemical substance, produced by a gland, carried by the blood, which alters the activity of one or more specific target organs and is then destroyed by the liver.

ADRENALINE

- A hormone secreted by the adrenal gland.
- It increases pulse rate, makes the glycogen in muscles get converted to glucose, and released into blood, makes you breathe deeper and more rapidly, airways become wider, and makes skin become pale as blood is diverted away.
- It increases the concentration of glucose in the blood for respiration.
- Adrenaline is secreted for example: while bungee jumping or riding a rollercoaster

NERVOUS AND HORMONAL SYSTEMS

Comparison	Nervous system	Endocrine system
Speed of action	Very rapid	Can be slow
Nature of message	Electrical impulses, travelling along nerves	Chemical messenger (hormones) travelling in bloodstream

<i>Duration of response</i>	Usually within seconds	May take years (puberty)
<i>Area of response</i>	Localized response (only one area usually)	Widespread response (in many organs)
<i>Example of process controlled</i>	Reflexes such as blinking	Development of reproductive system

- Hormones are used in food production, for example oestrogen is used to boost growth rate of chickens.
- Advantage: chickens grow quickly meaning more profit.
- Disadvantages: this may cause human males to develop feminine characteristics, and it is unnatural.

TROPIC RESPONSES

- Auxin:**
 - Plant hormones or growth substances
 - Controls tropisms
 - It is produced by cells at the tip of roots and shoots of plants
- Geotropism:** a response in which a plant grows towards (positive) or away (negative) from gravity.
- Auxins' role in geotropism:**
 - Tend to settle at the bottom end of the root.
 - However, this does not make the cells of the tip of the root grow longer; auxins prevent cells at bottom tip of root from growing, making cells at top of root grow faster.
 - When cells at top of the root grow faster, they push root deeper into soil and root gets longer.
 - The root grows in direction of the gravitational pull.
- Phototropism:** a response in which a plant grows towards (positive) or away (negative) from the direction from which light is coming.
- Auxins' role in phototropism:**
 - If sun shines on right side of a plant's shoot, auxins will accumulate on dark opposite left side.
 - Auxins accumulating makes cells on left side grow faster than cells on right side.
 - When left side of shoot starts growing faster than right side, shoot will start to bend to right side towards sunlight.
- Hormones can be used as weed killers:** spraying with high concentrations of hormone upsets normal growth patterns. It affects different species differently so might only kill one species not the other (this is good).

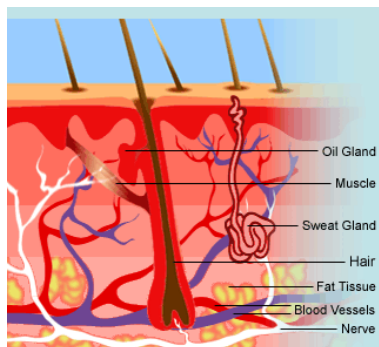
HOMEOSTASIS

- The maintenance of a constant internal environment.

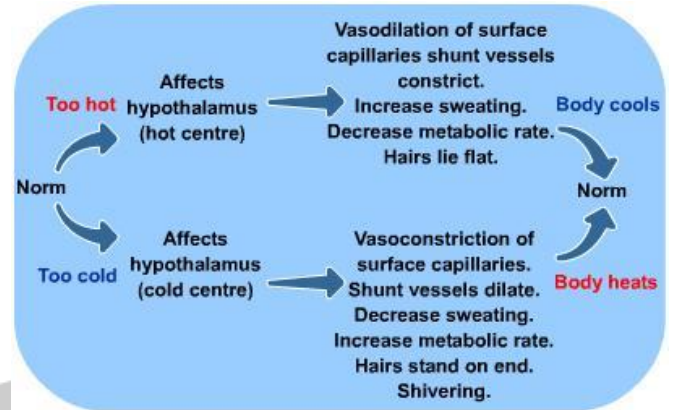
THERMOREGULATION

Constant body temperature is maintained by:

- Insulation:** provided by fatty tissue retains heat. Hairs become erect to trap warm air by contracting erector muscles and vice versa.
- Vasodilatation:** when it is hot, arterioles, which supply blood to the skin-surface capillaries, dilate (become wider) to allow more blood near to skin surface to increase heat loss (face redder)



- Vasoconstriction:** when it is cold, arterioles, which supply blood to the skin-surface capillaries, constrict (become smaller) to allow less blood near to skin surface to decrease heat loss
- Sweating:** the water evaporates giving a cooling effect
- Skin receptors:** sense heat and sensory neurons send impulses to the hypothalamus
- Shivering:** muscular activity generates heat
- Thermoregulatory center:** in the hypothalamus, it controls the use of corrective mechanisms (e.g. sweating and shivering).



OSMOREGULATION

- It is the body's way of balancing water taken in by the diet and water lost by excretion.
- This stops red blood cells from becoming crenated
- In the collecting duct, water is reabsorbed into the blood depending on how much is needed. This is controlled by the antidiuretic hormone, ADH released by the pituitary gland.
- If we sweat then the volume of urine is reduced to compensate
- Isotonic sports drinks are used by athletes because they have glucose, salts and water to replace what is lost by sweating

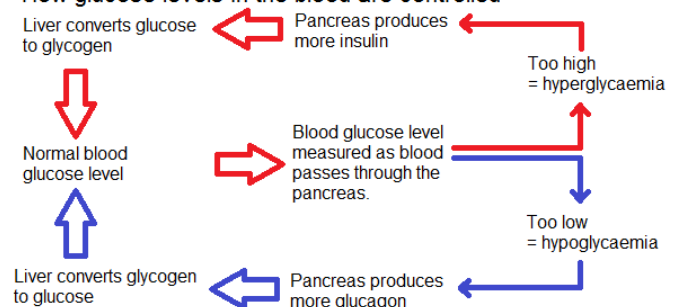
NEGATIVE FEEDBACK

- The production of hormones is controlled by feedback – the hormones regulate their own production.
- A negative feedback control is when the change in hormone level acts as a signal to cancel out that change, so when blood hormone level is low, hormone production is stimulated, when it is high, it is inhibited.

GLUCOREGULATION

- Blood glucose levels are monitored and controlled by the pancreas
- The pancreas produces and releases different hormones depending on the blood glucose level
- Insulin is released when blood glucose levels are high – the liver stores excess glucose as glycogen
- Glucagon is released when blood glucose levels are low – the liver converts stored glycogen into glucose and releases it into the blood

How glucose levels in the blood are controlled



HOMEOSTATIC ORGANS

- Cells: change composition of blood as they remove nutrients and O₂ and add wastes and CO₂
- Heart: keeps blood pressure constant to deliver oxygen and nutrients around body
- Skin: to maintain heat exchange with external environment
- Kidneys: regulate water and salt levels (osmoregulation) and the removal of wastes like urea (excretion)
- Lungs: regulate gas exchange
- Intestines: supply soluble nutrients and water to blood
- Liver: regulates blood solutes and removes toxins

DRUGS

- Any substance taken into the body that modifies or affects chemical reactions in the body.

ANTIBIOTICS:

- Antibiotics work by stopping a metabolic process performed by the bacteria you are trying to get rid of, but not performed by human cells.
- Antibiotics don't work on viruses because they are not really living and they make the host cell perform the tasks for them.

HEROIN

Effects of the abuse of heroin: a powerful depressant

- Problems of addiction
- Severe withdrawal symptoms (vomiting, restlessness)
- Malnourishment as drug depresses appetite
- Financial problems – stealing, loss of job
- Infection from sharing needles e.g. HIV/AIDS

ALCOHOL

Effects of excessive consumption of alcohol – a depressant:

- Causes coronary heart diseases
- Reduced self-control
- Depression
- Effect on reaction times
- Damage to liver – cirrhosis

SMOKING

Some effects of tobacco smoke:

- Drying effect and heat irritate lungs – destroys cilia
- Nicotine is addictive, it is also a stimulant, it increases pulse rate and narrows blood vessels which can cause damage
- Tar causes cancer, and is an irritant so causes coughing.
- There are other irritants in tobacco smoke including: smoke particles, ammonia, and sulfur dioxide
- Emphysema: walls between alveoli break making large sacs, reducing surface area massively and making you breathless after a couple of steps
- Loss of limbs due to poor circulation, CHD and lower sperm-count
- Carbon monoxide irreversibly bonds with haemoglobin which can lead to oxygen starvation
- Cancer of the stomach, pancreas and bladder etc.

REPRODUCTIONASEXUAL REPRODUCTION

- The process resulting in the production of genetically identical offspring from one parent.
- **Bacteria:**
 - Reproduce by binary fission, each bacterium divides into two.
 - The generation time is the time taken for a cell to divide into 2.

Fungi:

- Single-celled yeast reproduces by binary fission.
- All other fungi produce via spores.
- When the sporangium bursts it spreads the spores.
- Spores land and grow mycelium (roots) for example mushrooms

Potatoes:

- The shoot from a potato goes back underground and the stem swells to form a new genetically identical potato.
- The swollen stem acts as a storage organ.

Advantages

- Fast: no need to find mate, fertilise etc.
- Good characteristics are kept

Disadvantages

- No variation
- Harmful genes transferred
- Overcrowding- fighting for food

SEXUAL REPRODUCTION

- The process involving the fusion of haploid nuclei (23 chromosomes) to form a diploid zygote (46 chromosomes) and the production of genetically dissimilar offspring.

Advantages

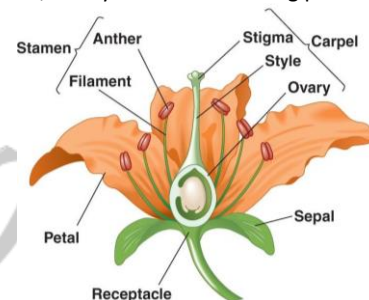
- Produces genetically different offspring; don't all die from change in the environment.

Disadvantages

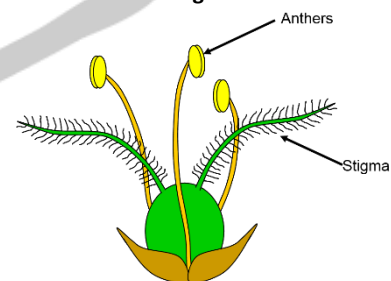
- Takes lots of time and energy
- Good characteristics can be lost
- Energy on improving appearances or pollen volume for pollination (plants)

SEXUAL REPRODUCTION IN PLANTS

- Insect pollinated, dicotyledonous flowering plant: **foxglove**



- Wind pollinated flower structure: **grass**

FUNCTIONS

- **Sepal:** protect the flower bud.
- **Petal:** brightly colored and scented and may have nectaries which are all used to attract insects, petals in wind pollinated flowers are tiny, and used for pushing the bracts (leaf-like structures) apart to expose stamens and stigma
- **Anther:** has pollen sacs with pollen grains which contain the male nucleus (male gamete).
- **Stigma:** platform on which pollen grains land
- **Ovary:** hollow chamber, ovules grow from the walls.

POLLINATION

- The transfer of pollen grains from the male part of the plant (anther of stamen) to the female part of the plant (stigma).
- **Agents of pollination:** insects, birds, mammals, water and wind

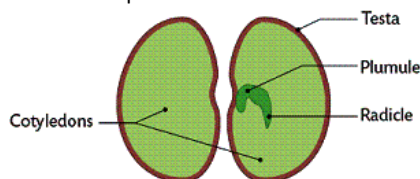
Insect Pollinated

- Large colourful petals – attract
- Sweetly scented
- Nectaries
- Moderate amount of pollen
- Pollen is spiky/sticky
- Anther & stigma inside flower
- Stick stigma
- Flowers have stripes which act as guide-lines for insects

Wind Pollinated

- Dull petals
- No scent
- No nectaries
- Huge amount of pollen
- Pollen round and smooth
- Anther & stigma hangs out
- Stigma hairy

- **Pollen tube:** pollen grain lands on stigma and creates a tunnel down the style, through the micropyle, to the ovules.
- Structure of non-endospermic seed:



Plumule + Radicle = embryo

- **Formation of a seed:** the zygote divides many times by mitosis to form an embryo. The cotyledon is the food store. The testa stops drying out of embryo.
- Wind and animal dispersal are used by plants to colonise new areas; done because new areas have less competition for light, space and nutrients, so seeds are more likely to develop.

Wind Dispersed Seed

- Dandelion



- Sycamore

**Animal Dispersed Seed**

- Apple (internal)



- Bur (external)

**Self Pollination**

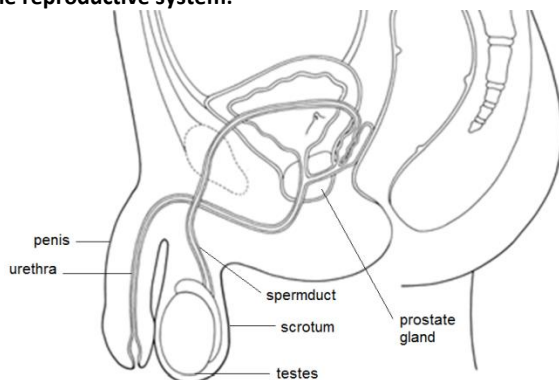
- Pollen is transferred from the anther to the stigma of the same flower.
- Implications:
 - Very efficient
 - No genetic variation

Cross Pollination

- Pollen transfer from anther to stigma of another flower of the same species.
- Implications:
 - Risky: pollen might not reach the other flower
 - Chance for genetic variation

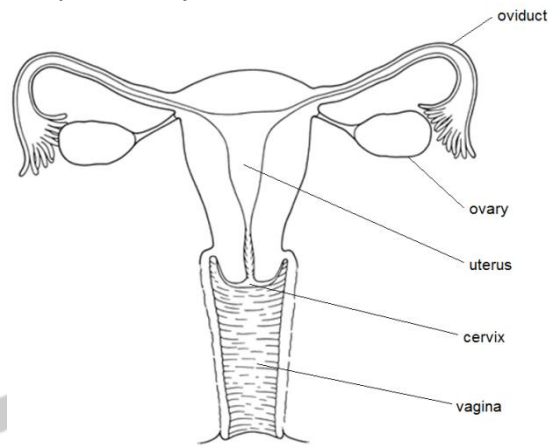
SEXUAL REPRODUCTION IN HUMANS

- **Male reproductive system:**

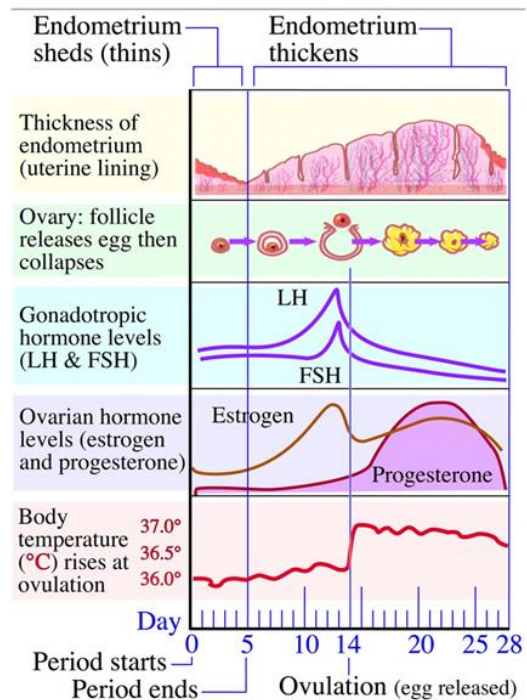


- **Testes:** have many coiled tubes which produce sperm, and the cells between tubes produce testosterone.
- **Scrotum:** holds testicles
- **Sperm duct:** carries sperm from testicles to urethra.
- **Prostate gland:** makes seminal fluid
- **Urethra:** carries semen from sperm duct to tip of penis
- **Penis:** male sex organ, used to transfer semen to the female.

- **Female reproductive system:**



- **Ovary:** contains follicles which develop into the ova and produces progesterone and oestrogen
- **Oviduct (fallopian tube):** carries the ovum to the uterus
- **Uterus (womb):** where the fetus develops.
- **Cervix:** neck of uterus: a strong rigid muscle, moist by mucus with a small opening
- **Vagina:** receives penis during intercourse, and way out for baby at birth. Moist tube of muscle, flexible and secretes mucus

MENSTRUAL CYCLE

- **Day 1 to 5:**

- In the ovary, **FSH** secreted by the **Pituitary Gland** to stimulate the maturation of **ONE follicle** in the ovary.
- In the uterus: the **endometrium** breaks down; menstruation

- **Day 5 to 12:**
 - In the ovary the follicle keeps maturing
 - In the uterus, oestrogen is secreted by follicle and the ovarian tissues to prepare the endometrium
- **Day 13/14/15:**
 - In the ovary, LH is also secreted by the Pituitary Gland to trigger the release of the egg from follicle into the fallopian tube
- **Day 15 to 28:**
 - In the ovary, LH triggers the formation of the Corpus Luteum
 - In the uterus: progesterone is secreted by Corpus Luteum to keep endometrium thick, waiting for possible embryo implant.
- **Day 28 – Scenario 1:** Egg not fertilized
 - No implantation takes place, the Corpus Luteum degenerates, causing a lack of progesterone.
 - This means that endometrium is no longer thick, back to Day 1
- **Day 28 – Scenario 2:** Egg is fertilized
 - Implantation occurs.
 - This makes the hormones to keep the Corpus Luteum maintained which means that progesterone is high.
 - This keeps the Endometrium thick for pregnancy.

HORMONES IN MENSTRUAL CYCLE

- **Oestrogen** is secreted by the ovaries. It stops FSH being produced - so that only one egg matures in a cycle and it stimulates the pituitary gland to release the hormone LH.
- **Progesterone** is a hormone secreted by ovaries. It maintains the lining of the uterus during the middle part of the menstrual cycle and during pregnancy.
- **Follicle stimulating hormone (FSH)** is secreted by the pituitary gland. It causes an egg to mature in an ovary and it stimulates the ovaries to release the hormone oestrogen
- **Luteinizing hormone (LH):** is also secreted by the pituitary gland and causes the mature egg to be released from the ovary.

SEXUAL INTERCOURSE

- Penis fills with blood and becomes erect
- Vagina walls secrete a lubricant.
- Rubbing of the glans (end of penis) against the vagina wall sets off a reflex action, causes sperm to be released from the testes, and is transported by peristalsis along sperm ducts and urethra, where seminal fluid is added to make semen.
- The exit of semen from the penis is called ejaculation.
- Sperm then swim through the cervix and oviducts to the first third of the oviduct (from the ovary) where one combines with the egg.

FERTILIZATION

- The fusion of an ovum and a sperm to form a zygote.
- **Development of zygote:**
 - One sperm penetrates
 - The ovum membrane alters to form a barrier against sperm.
 - Head of sperm (male nucleus) approaches and then fuses with the nucleus of the ovum.
 - Zygote divides over and over, to make a ball of cells called an embryo.
 - It implants itself in the wall of the nucleus (implantation) which is followed by conception
- **Development of fetus:** zygote is changed through growth (mitosis) and development (organization of cells into tissues and organs)
- **Umbilical cord:** contains umbilical artery which carries deoxygenated blood and waste products from fetus to placenta and umbilical vein which carries oxygenated blood and soluble food from placenta to fetus. (Contains fetus' blood)

- **Placenta:** organ for exchange of soluble materials such as foods, wastes and oxygen between mother and fetus; physical attachment between uterus and fetus. (Contains mother's blood)
- **Amniotic sac:** membrane which encloses amniotic fluid, broken at birth.
- **Amniotic fluid:** protects fetus against mechanical shock, drying out and temperature fluctuations.

ANTENATAL CARE:

- **Change in diet:**
 - More proteins → growth of fetus
 - Slightly more fat → the new cells' cell membrane
 - More vitamin C and D → blood vessel walls and bones
 - Iron → haemoglobin
 - Calcium → growth of bones and teeth
- Guidance on motherhood
- Checks on fetus and mother including: weight check, blood tests, urine tests, blood pressure checks, ultrasound scanning etc.

LABOR AND BIRTH

- **Labor:** The uterine muscular wall contract and cervix tries to relax, then contractions get more frequent. Contractions cause amniotic membrane to break and release amniotic fluid.
- **Expulsion:** Powerful Contraction pushes baby out.
- **Afterbirth:** Placenta is expelled out. All contraction & pain gone

Gamete	Size	Mobility	Number
Sperm	Smaller	Very mobile – use its tail	Many more (300,000,000)
Egg	Larger	Immobile – moved by peristalsis	Fewer and limited

	Breast feeding	Bottled milk
Advantages	• Has antibodies – no bacteria • Nutrients- correct proportion • Correct temp. • No additives/preservatives • Builds mother-child bond • No cost/preparation • Causes decline in uterus size	• Less painful • Other people can feed baby • May contain supplement vitamins and minerals
Disadvantages	• May be painful • Mother must be present • Damage beauty	• More likely to develop illness • Risk of wrong mixture • Expensive

SEX HORMONES

- At puberty, the pituitary gland starts to stimulate the primary sex organs; the testes in males and the ovaries in females.
- Sex hormones – testosterone in males and oestrogen in females are released into the bloodstream.
- They only affect the target organs which have receptors which can recognize them.
- Causes secondary sexual characteristics such as the growth of pubic hair and maturation of sexual organs.

METHODS OF BIRTH CONTROL

- **Natural:**
 - Abstinence: don't have sex
 - Rhythm method: don't have sex during the fertile period, only during the safe period

• Chemical:

- Progesterone-only pill: pill which affects the uterus and makes implantation difficult
- Spermicide: a chemical applied as a gel, cream or foam which kills sperm. It is very unreliable on its own but makes barrier methods of contraception more effective.

• Mechanical:

- Condom: thin rubber covering over penis, it protects from impregnation and STDs, used by man
- Diaphragm: used by woman, prevent sperm entering uterus, reliable, must stay in place 6 hours after sex, needs a correct size
- Femidom: closed end, has a ring which gets pushed through cervix and open end's ring lies against the labia
- IUD: plastic-coated copper coil, can be left inside for months or even years, has a string which is used to remove it out of the vagina, reliable, it irritates uterus wall preventing implantation

• Surgical:

- Vasectomy: sperm ducts are cut and tied, 100% reliable
- Female sterilization: oviducts are cut and tied, 100% reliable

ARTIFICIAL INSEMINATION

- **By donor:** man's sperm has a problem, making impregnation impossible, so a donor gives his sperm.
- **In vitro fertilization:** an ovum is fertilized outside a woman's body. The fertilized ovum is implanted into the uterus.
- **Fertility drugs:** drugs which enhance reproductive fertility. For women, fertility medication is used to stimulate follicle development of the ovary. The side effect is multiple pregnancies. They contain varying amount of FSH and LH.

GONORRHEA

- A bacterial infection caused by penetrative sex through the mouth, vagina or anus
- **Symptoms/signs:**
 - Pain or burning when passing urine
 - Greenish/yellow discharge from the penis or vagina
 - Inflammation of the testicles
- **Effects:**
 - In men the urethra becomes infected, in woman it is the cervix.
 - If left untreated, the disease can travel through reproductive tract causing sterility
- **Treatment:**
 - Can be cured with penicillin however no immunity

HUMAN IMMUNODEFICIENCY VIRUS (HIV)

- **Transmission:** Intercourse, blood transfusion, organ transplant or sharing needle with infected person
- **Prevention:**
 - Avoid intercourse with many partners
 - Use a condom
 - Don't come in contact with other people's blood
- **How it affects the immune system:**
 - Infects and destroys lymphocytes
 - Decreases efficiency of immune system
 - Body becomes liable to infection by other pathogens

GROWTH AND DEVELOPMENT

- **Growth:** a permanent increase in size and dry mass by an increase in cell number or cell size or both
- **Development:** increase in complexity.

GERMINATION

- A process controlled by enzymes
- **Water:** activates enzymes to turn insoluble food stores into soluble substances, and makes tissues swell so that the testa splits
- **Oxygen:** enters through the gaps in the testa (along with water), and is used in aerobic respiration.
- **Temperature:** must be suitable for enzymes to work (at optimum temperature).

INHERITANCE

- The transmission of genetic information from generation to generation.

CHROMOSOME

- **Chromosome:** a thread of DNA, made up of a string of genes
- **Gene:** a length of DNA that is the unit of heredity and codes for a specific protein. A gene may be copied and passed on to the next generation
- **Allele:** any of two or more alternative forms of a gene
- **Haploid nucleus:** a nucleus containing a single set of unpaired chromosomes (e.g. sperm and egg)
- **Diploid nucleus:** a nucleus containing two sets of chromosomes (e.g. in body cells)
- **Inheritance of gender in humans:** woman's gamete can only carry an "X" chromosome, and a male gamete can carry either an "X" or "Y" chromosome; females are "XX" while males are "XY". There is always a 50% chance of getting a boy and vice versa.

MITOSIS

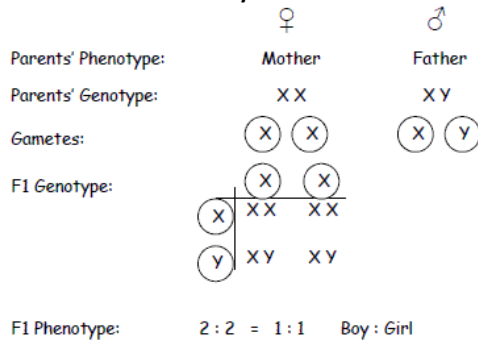
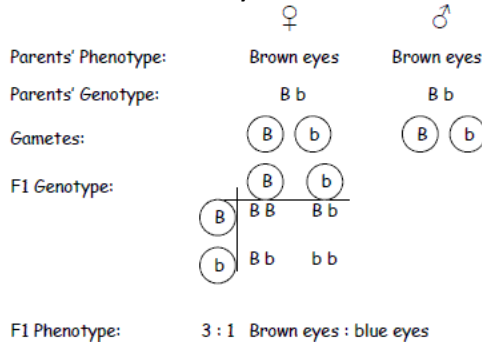
- The nuclear division giving rise to genetically identical cells in which the chromosome number is maintained by the exact duplication of chromosomes.
- Mitosis is needed for:
 - **Growth:** in animals each tissue provides its own new cells when they are needed.
 - **Repair** of damaged tissues: for example when you cut your skin, mitosis provides the new cells to cover up the cut.
 - **Replacement** of worn out cells
 - **Asexual reproduction:** in plants

MEIOSIS

- Reduction division in which the chromosome number is halved from diploid to haploid
- Gametes are the result of meiosis
- Meiosis results in genetic variation so the cells produced are not all genetically identical.

MONOHYBRID INHERITANCE

- **Genotype:** genetic makeup of an organism in terms of the alleles present (e.g. Tt or GG)
- **Phenotype:** physical or other features of an organism due to both its genotype and its environment (e.g. tall plant or green seed)
genotype + environment + random variation → phenotype
- **Homozygous:** having two identical alleles of a particular gene (e.g. TT or gg). Two identical homozygous individuals that breed together will be pure-breeding
- **Heterozygous:** having two different alleles of a particular gene (e.g. Tt or Gg), not pure-breeding
- **Dominant:** an allele that is expressed if it is present (e.g. T or G)
- **Recessive:** an allele that is only expressed when there is no dominant allele of the gene present (e.g. t or g)

1:1 Monohybrid Crosses**3:1 Monohybrid Crosses**

Note the gametes are always put in circles

- **Co-dominance:** when neither of two alleles is dominant to each other.
- There are three alleles for blood group given by the symbols I^A , I^B and I^O .
- I^A and I^B are co-dominant giving blood group AB or $I^A I^B$, and both dominant to I^O .

VARIATION

- **Continuous variation** is influenced by genes and environment, resulting in a range of phenotypes between two extremes, e.g. height in humans
- **Discontinuous variation** (e.g. you are either blood group O, A, B or AB, nothing else) is caused by genes alone and results in a limited number of distinct phenotypes
- **Mutation:** a change in a gene or chromosome
- Mutation is a source of variation e.g. in Down's syndrome, where a parent's chromosomes are unevenly distributed in meiosis. In fertilisation, a zygote with a number of chromosomes that is not 46 is created (e.g. 23 + 24). Characteristics: broad forehead, short neck, downward-sloping eyes, short nose and mental retardation.

SICKLE CELL ANAEMIA

- Disease in which the red blood cell has a sickle shape instead of a round biconcave shape, controlled by a recessive allele, which causes weakness, aching joints and poor circulation.
- The fact that it is recessive means that a heterozygous person can be a carrier: they have the allele but it is not expressed.
- Being a carrier of sickle cell anaemia makes you resistant to malaria
- In equatorial Africa, being sickle cell anaemic causes death, malaria causes death, but the carriers have immunity to malaria and have some symptoms of anaemia, in severe cases they are very weak.

MUTAGENS

Rate of mutation increases with:

- **Chemicals:** tars in tobacco smoke, high concentrations of some preservatives and some plant control hormones
- **Radiation:** gamma, ultraviolet and X-radiation can damage and cause mutations because they have an ionising

ARTIFICIAL SELECTION

- Is breeding organisms with valued characteristics together in order to try to produce offspring which shares those useful characteristics (selective breeding).
- Can be used to produce organisms which are more economically valued
- For example: cows that produce more milk, wheat that is easier to separate from grain, dogs which have better appearance

NATURAL SELECTION

- The greater chance of passing on of genes by the best adapted organisms.
- Variation is natural or random changes in all living organisms.
- Variation leads to survival of the fittest since the variations in certain organisms allow that organism to have an advantage over the others in its species in that area
- The surviving organisms reproduce, since they don't get eaten up, so the variation has caused the species to evolve.
- Evolution is caused by natural selection which is caused by a change in the environment.

RESISTANT ANTI-BIOTIC

- Strains of antibiotic-resistant bacteria are developing as the use of antibiotics is increasing.
- In a group of many bacteria, one might mutate to be resistant to the antibiotic, as a result it reproduces and the others die making a new strain of bacteria, which is resistant to antibiotics.

GENETIC ENGINEERING

- Taking a gene from one species and putting it into another species.

Human Insulin in Bacteria

- The gene coding from a pancreas cell for the production of human insulin is 'cut' from chromosome fragments.
- The plasmid (DNA) from a harmless bacteria cell is cut to remove a part and they are combined to form a recombinant DNA.
- The bacteria are put in a fermenter or bioreactor to get a large population, and then the product is processed.

ENERGY FLOW

- The sun is the principal source of energy input to biological systems.
- Energy flow is **not** a cycle; it starts from the sun and then that energy is harnessed by plants which are eaten by animals which are eaten by other animals.
- At each step, energy is lost to the environment.

FOOD CHAINS AND FOOD WEBS

- **Food chain:** a chart showing the flow of energy (food) from one organism to the next beginning with a producer, for example:
mahogany tree → caterpillar → song bird → hawk
- **Food web:** a network of interconnected food chains showing the energy flow through part of an ecosystem
- **Producer:** an organism that makes its own organic nutrients, usually using energy from sunlight, through photosynthesis
- **Consumer:** an organism that gets its energy by feeding on other organisms

- **Herbivore:** an animal that gets its energy by eating plants
- **Carnivore:** an animal that gets its energy by eating other animals
- **Decomposer:** an organism that gets its energy from dead or waste organic matter (i.e. a saprotroph)
- **Ecosystem:** a unit containing all of the organisms and their environment, interacting together, in a given area e.g. decomposing log or a lake
- **Trophic level:** the position of an organism in a food chain, food web or pyramid of biomass, numbers or energy

Food chains usually have fewer than five trophic levels, because energy transfer is inefficient:

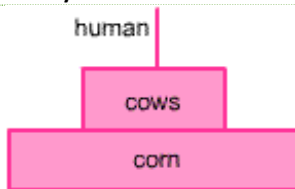
- Sun produces light, less than 1% of the energy falls onto leaves.
- Producers 'fix' only about 5-8% of that energy, because of: transmission, reflection and incorrect wavelength.
- Primary consumer only gets between 5-10% because some parts are indigestible (e.g. cellulose) and not eating the whole plant.
- Secondary consumer gets between 10-20% because animal matter is more digestible and has a higher energy value.
- At each level heat is lost by respiration.

Humans eating plants is more efficient than humans eating animals because...

- We need only a couple of vegetables to have one meal, but to have meat we must feed the animal a lot of plant material in order to get far less meat.
- In the process of raising an animal, plants lose energy to environment, then animal loses energy to environment and does not use up all the plant material so it is very inefficient.

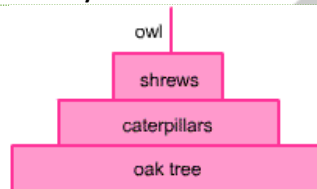
PYRAMID

Pyramid of Numbers



- Shows number of each organism in a food chain.
- When moving up pyramid, number of individuals' decreases

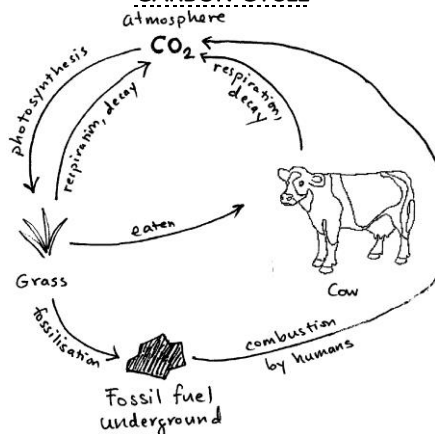
Pyramid of Biomass



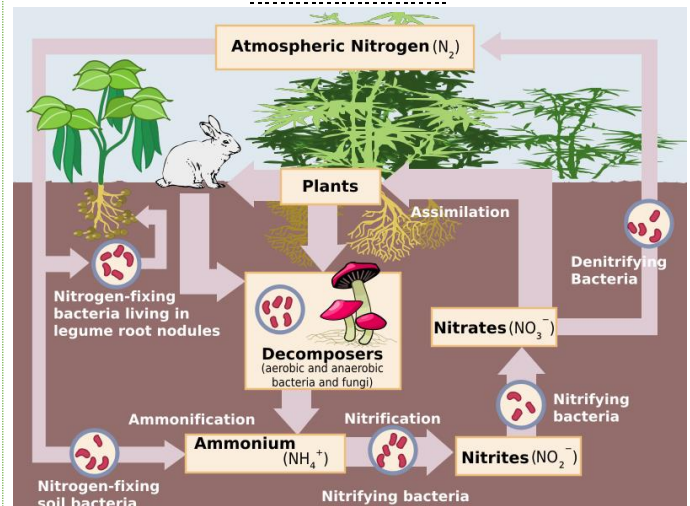
- Pyramid which shows the biomass
- (number of individuals × their individual mass)

NUTRIENT CYCLES

CARBON CYCLE

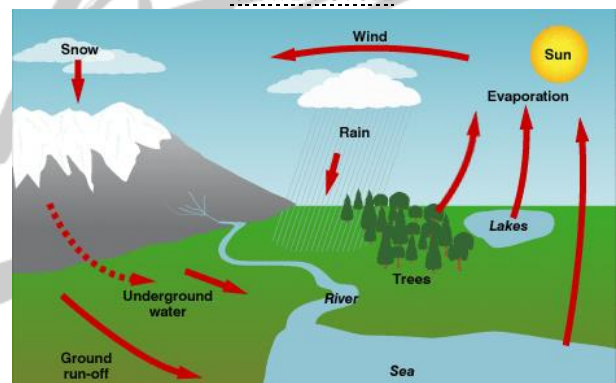


NITROGEN CYCLE



- Nitrogen-fixing bacteria provide usable nitrogen for plants, these may exist in the root nodules where they live in symbiosis with the plants (nitrogen fixation), or this can happen because of lightning, or microorganisms provide them through decomposition.
- Nitrifying bacteria convert nitrogen-containing substances into better nitrogen-containing substances for the plants (nitrification).
- Plants absorb these substances and convert them into proteins
- Primary consumers eat the plants and can make their own proteins, secondary consumers eat primary consumers and so on.
- Death and decay happens at each trophic level leading to stage one
- Denitrifying bacteria carry out denitrification: they convert nitrogen-containing substances into atmospheric nitrogen

WATER CYCLE



	O ₂ conc. in air	CO ₂ conc. in air	Why?
Combustion of fossil fuels	Decreases	Increases	• Burning uses up oxygen, and produces carbon dioxide
Deforestation	Decreases	Increases	• Fewer trees means less photosynthesis • Trees are usually burnt (combustion) • Decomposition of tree trunks (respiration)

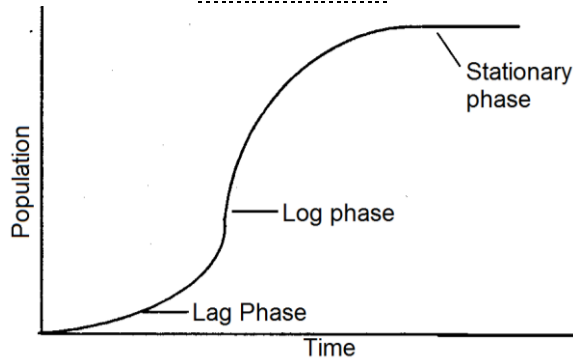
POPULATION

- **Population:** a group of organisms of one species, living in the same area at the same time

FACTORS AFFECTING RATE OF POPULATION GROWTH:

- **Food supply:** quantity and quality, for example snails need calcium to reproduce to make a shell (food quality).
- **Predation:** if predator population falls, the prey population will rise
- **Disease:** causes organisms to die so a high death rate partly cancels out birth rate meaning less population growth, especially if the organism dies before giving birth, or even population decline

SIGMOID CURVE



- **Lag phase:** number of mature, reproducing individuals is low and they may be widely dispersed
- **Log phase:** exponential growth occurs, the conditions are ideal and maximum growth rate is reached. Limiting factors do not limit growth much.
- **Stationary phase:** limiting factors slow growth as population has reached "carrying capacity" of its environment; mortality rate = birth rate; curve levels off and fluctuates around this maximum population size.

HUMAN POPULATION GROWTH

Factors favouring growth:

lower infant mortality, higher life expectancy, better nutrition, better housing, better sanitation, medicine, vaccination

Factors controlling growth:

disease, famine, war

- The human population is **becoming stable (stagnation)** due to:
 - better education (particularly for women), so they work instead of getting married and having children
 - better living conditions, fewer people die, fewer births needed
 - cities, reduced need for physical labour on farms
 - family planning
- but overall the population is still increasing.
- **Social implications** of human growth:
 - demands for roads as there is an increase in number of cars
 - greater expectation for a variety of foods all year round
 - smaller families increase demand for housing
 - greater demand for leisure and recreation space

HUMAN INFLUENCES ON THE ECOSYSTEM

AGRICULTURE

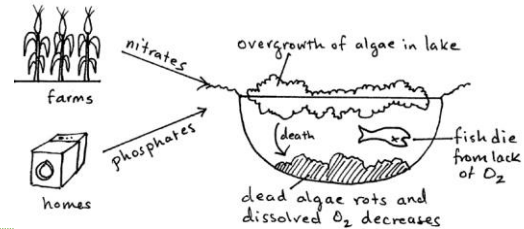
Deforestation:

- Reduced biodiversity/destroys habitats/extinction
- Loss of CO₂ fixation, thus increase in CO₂, thus global warming
- Soil erosion: tree roots cannot retain soil, goes into rivers making the water dirty & causes blockages, soil becomes less fertile

- **Flooding:** usually 75% of water is absorbed by foliage, root systems or evaporates. After deforestation water, accumulates in valleys

Eutrophication: when water plants receive too many nutrients.

- Fertilisers put in soil by farmers
- Fertilisers with nitrates / detergents with phosphates leach into rivers and lakes after rain
- Water plants grow more than usual
- They block sunlight and kill plants underneath
- They die and sink to bottom
- Bacteria/fungi decompose remains using the O₂ and decreasing the O₂ concentration
- Fish and other creatures die from oxygen starvation



POLLUTION

Water and air pollution:

- Chemical waste and sewage in rivers results in water not being drinkable and eutrophication can occur
- Sulfur dioxide dissolves in rain, causing acid rain which increases acidity of lakes and leaches aluminum out of the soil causing:
 - The fishes' gills are damaged, eventually killing them. This is fixed by adding calcium hydroxide (slaked lime)
 - Destroys the top of the trees and the aluminum damages tree roots = dead tree, important nutrients leached away
 - SO₂ poses health hazards for humans (asthma sufferers)
 - Damages limestone buildings and sculptures
 - Fewer crops can be grown on an acidic field (fixed by adding lime)

Pollution due to pesticides:

- **Insecticides (kill insects):** meant to kill insects which eat crops, but can kill other, useful insects such as bees which are pollinators, or by bioaccumulation (the increase in dose of toxin from one level of the food chain to the next)
- **Herbicides (kill weeds):** can be harmful to animals which eat the plants

Nuclear fall-out:

- Radioactive particles are sprayed into the atmosphere in a nuclear accident or bombing;
- These particles "rain" back to earth from clouds, sometimes far from the accident site;
- The radioactivity damages DNA and causes cancer and radiation illness at every level of the food chain.

Non-biodegradable plastics:

- Choke birds, fish and other animals
- Fill up the animals' stomachs so that they can't eat food
- Collect in rivers, and get in the way of fish

Acid rain:

- Caused by sulfur dioxide (burning fossil fuels) and nitrogen oxides (nitrogen reacting hot engines), as they dissolve and cause acid rain
- Damages trees and plants, and kills fish and other river life
- Prevention: catalytic converters, in factories slaked lime neutralizes these acidic oxides and use of flue-gas desulfurization

Global Warming:

- Increase in average temperature of the Earth
- Started at the same time as humans began burning fossil fuels
- Scientists believe fossil fuels are causing this – not proven yet

CONSERVATION

Species and habitats: need to be conserved because:

- Organisms have value in themselves (ethical value)
- Value to medicine (new molecules from exotic plants = new drugs)
- Genetic resources are useful to humans as well and are lost when species disappear (DNA for genetic engineering)
- Each species has its role in its ecosystem; if it is removed, then the whole ecosystem could collapse

Natural resources:

- **Water:** used to grow food, keep clean, provide power, control fires and to drink. We get water constantly through rainfall but we are using up planet's fresh water faster than it can be replenished.
- **Fossil fuels:** need to be conserved as they will soon run out, they should be therefore replaced with green forms of energy.

Recycling:

- **Water:** water from sewage can be returned to environment for human use by sanitation and sewage treatment
- **Paper:** sent to special centers where it is pulped to make raw materials for industry

NOTES